

ADC/DAC 技术

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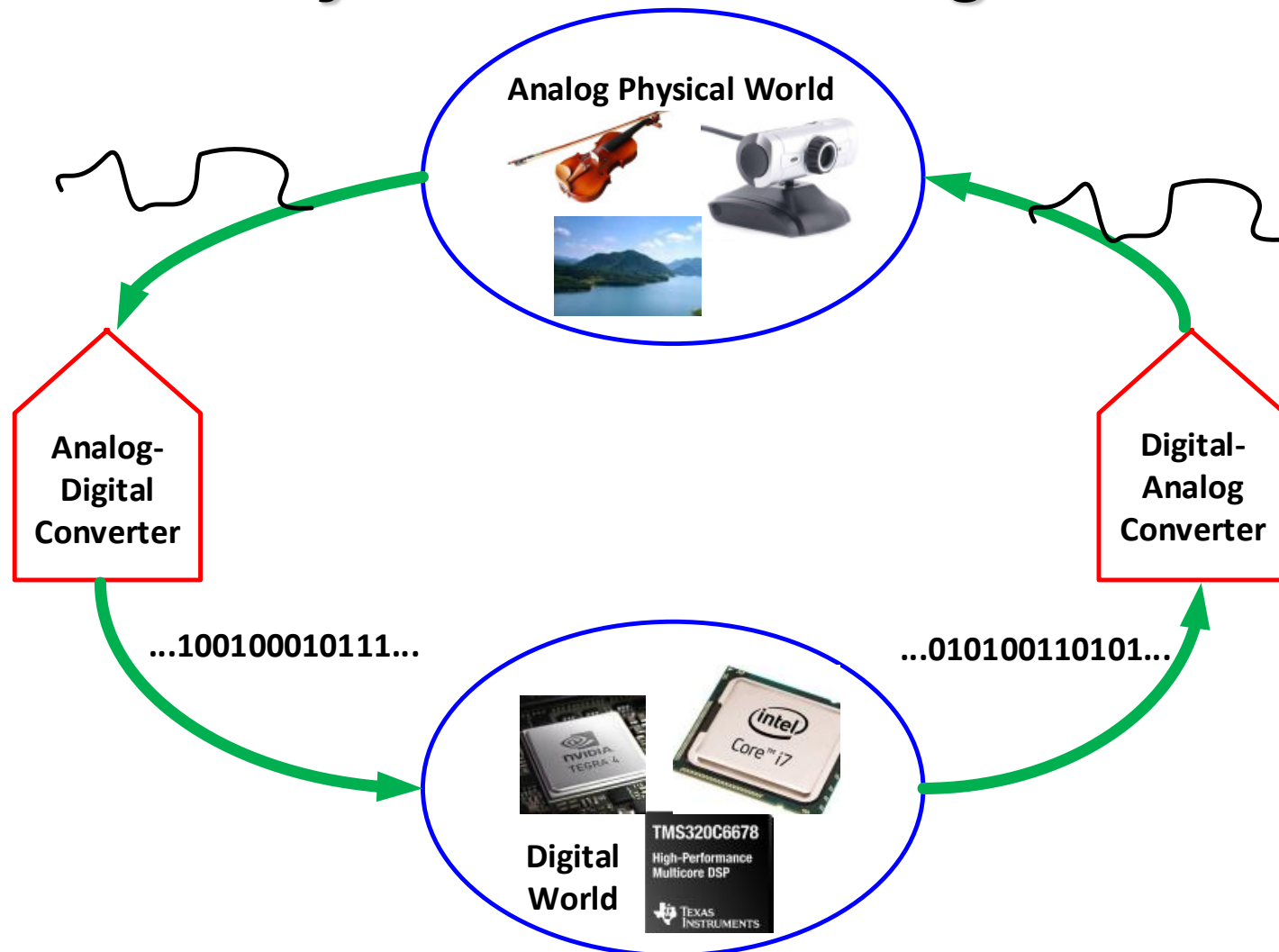




Outline

- **Application of ADCs/DACs**
- ADC/DAC Performance Spec.
- ADC Architectures
- State-of-the-Art ADCs
- State-of-the-Art DACs
- Industry High-Performance ADCs/DACs

Data Converters Serve as the Bridge between Physical World & Digital World

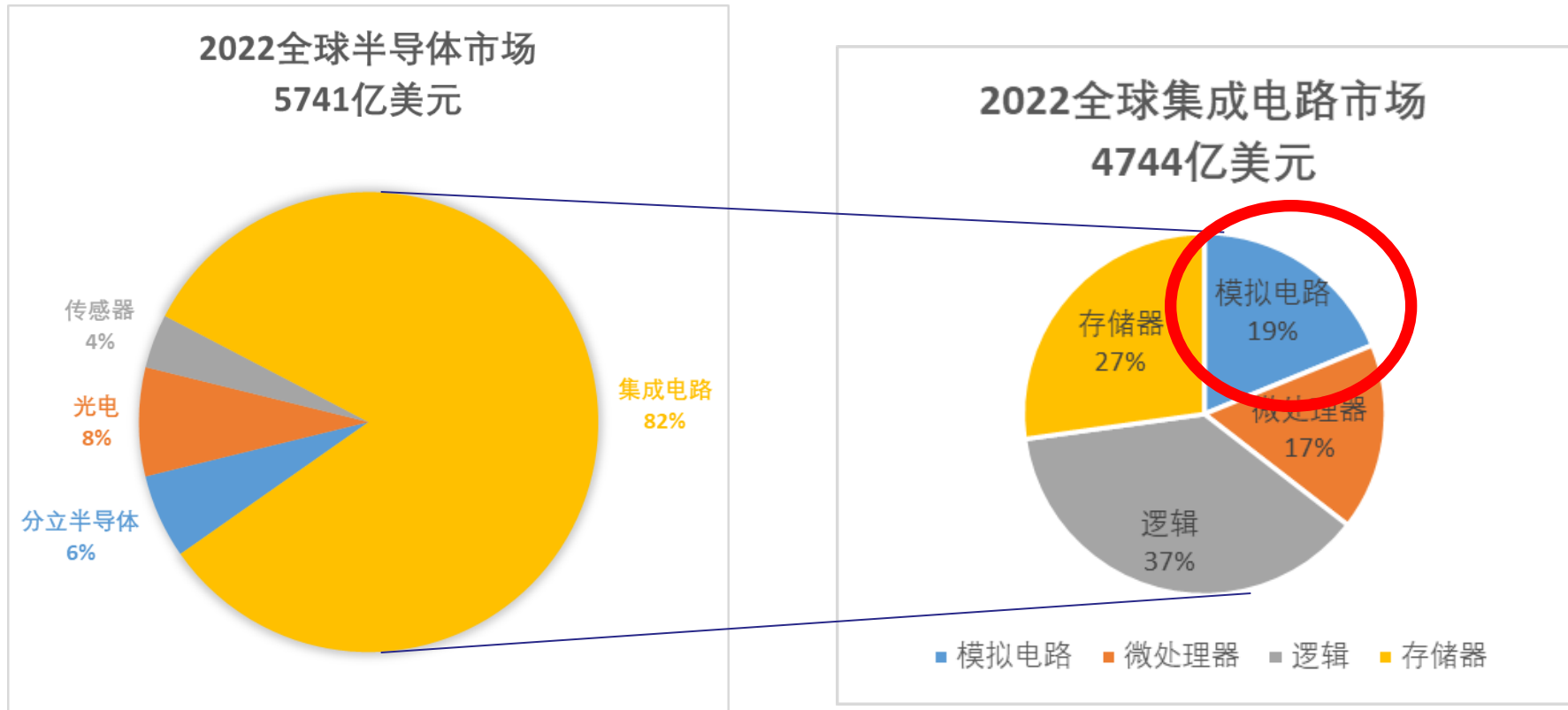


Data Converters Serve as the Bridge between Physical World & Digital World

- The physical world is analog
- The circuit world is ruled by the digital circuits
 - Robust: not sensitive to noise and disturbance
 - Programmable: function defined by software
 - Transplantable: easily transferred between platforms and extendible
 - Scalable with the development of CMOS technologies
- Analog-Digital Converters (ADC's) and Digital-Analog Converters (DAC's) are ALWAYS required to bridge the physical world and the digital world

Global IC Market

- ❑ 2024 global semiconductor market sales: \$627.6B
- ❑ 2022 global semiconductor market sales: \$474.4B, Analog IC: 19%



WSTS



2024 Global Analog IC Market \$87.5B





Analog IC Market Breakdown

2022 Analog IC Sales Forecast

	Market (\$M)	% of Total Analog	22/21 % Chg
General Purpose Analog			
Amplifiers & Comparators	4,481	5.4%	7%
Interface	3,030	3.6%	8%
Power Management	21,201	25.5%	12%
Signal Conversion	4,205	5.1%	8%
Total General Purpose Analog	32,917	39.6%	10%
Application-Specific Analog			
Consumer	3,106	3.7%	9%
Computer	3,048	3.7%	9%
Communications	26,233	31.5%	14%
Automotive	13,775	16.6%	17%
Industrial/Other	4,135	5.0%	9%
Total Application-Specific Analog	50,296	60.4%	13%
Total Analog Market	83,213	100.0%	12%

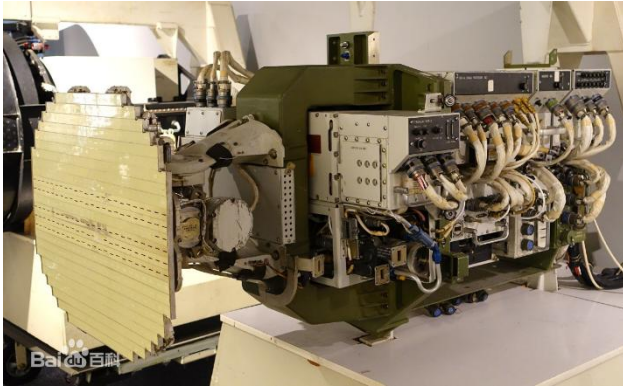
Source: IC Insights



Applications of Data Converters

- Instrumentation
 - Oscilloscopes
 - Signal generators
- Entertainment
 - Video
 - Audio
- Communications
 - Mobile communication: base stations/handsets
 - Wireless connections
- Military applications
 - Radar
- Medical applications
 - Sensor readout
 - Medical robots control
- Automotive electronics
 - Sensing
 - Actuator driver

Application Example—High Performance Radar



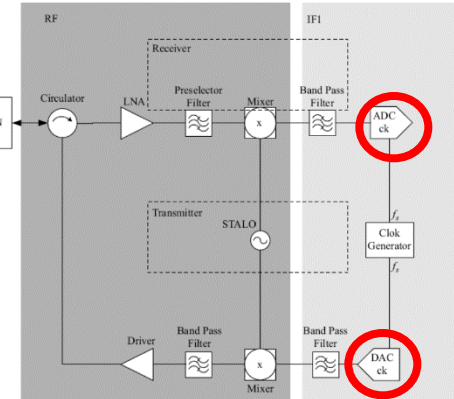
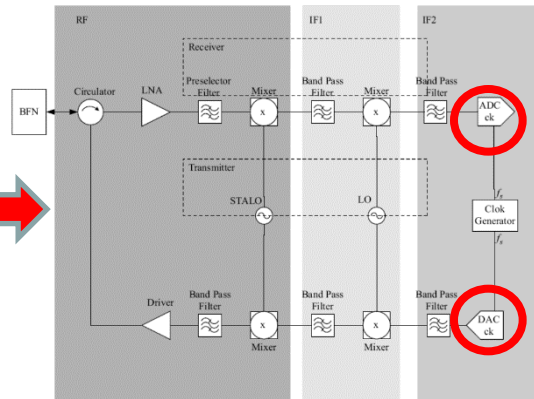
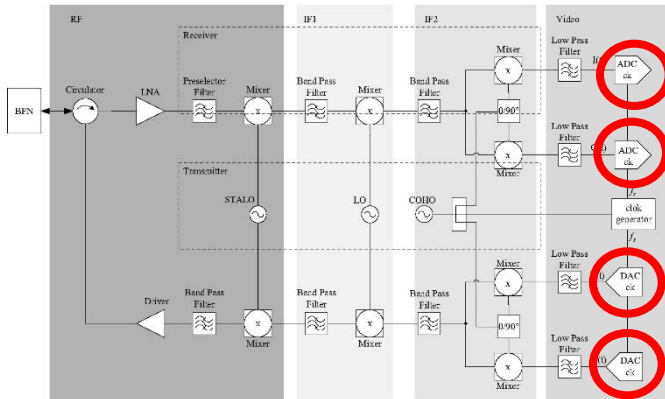
AN/APG-66 PD radar
ADC sampling rate 4MHz



AN/APG-81 AESA radar in F35
ADC sampling rate ~100MHz

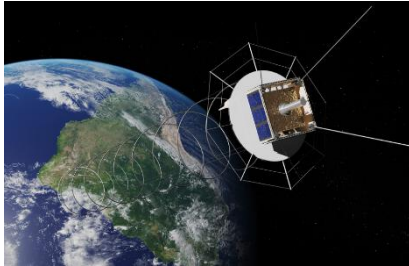


How about the newest
14bit/10Gbps RF ADC?

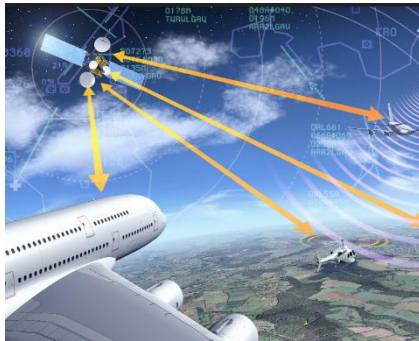


The continuously improved sampling rate of ADC/DAC is the key to reduce radar size while improving the performance

Application Example—High Speed Communication



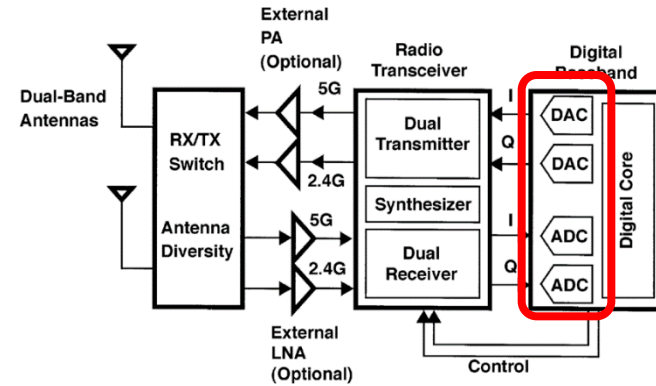
Satellite communication



Airborne communication

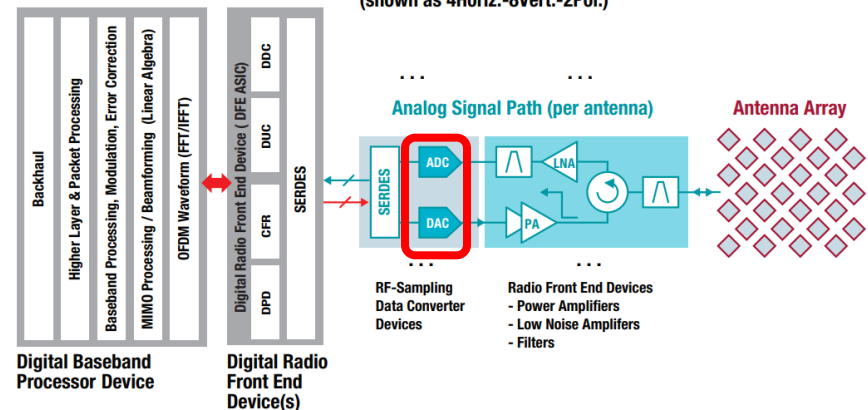


Base station



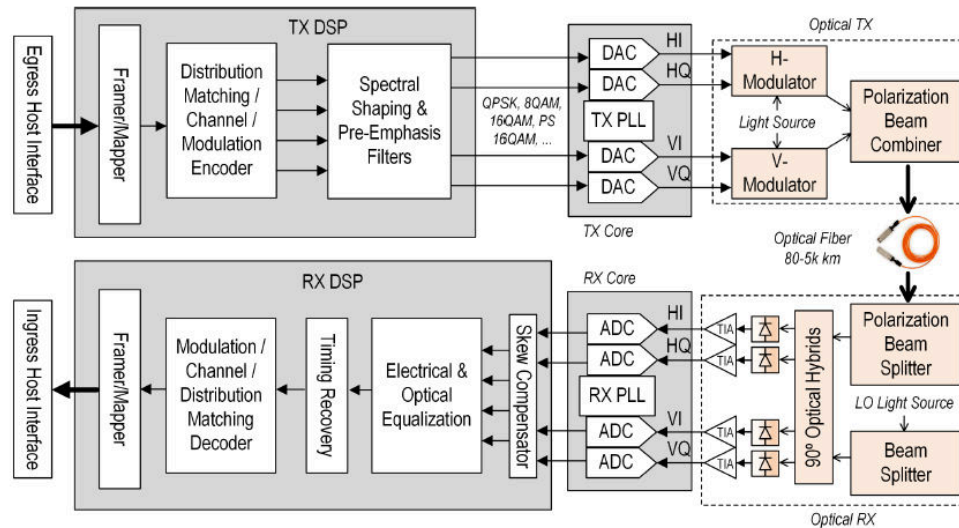
- ADC/DAC is the key component in transceivers

Typical 64 Antenna MIMO Base Station
(shown as 4Horiz.-8Vert.-2Pol.)

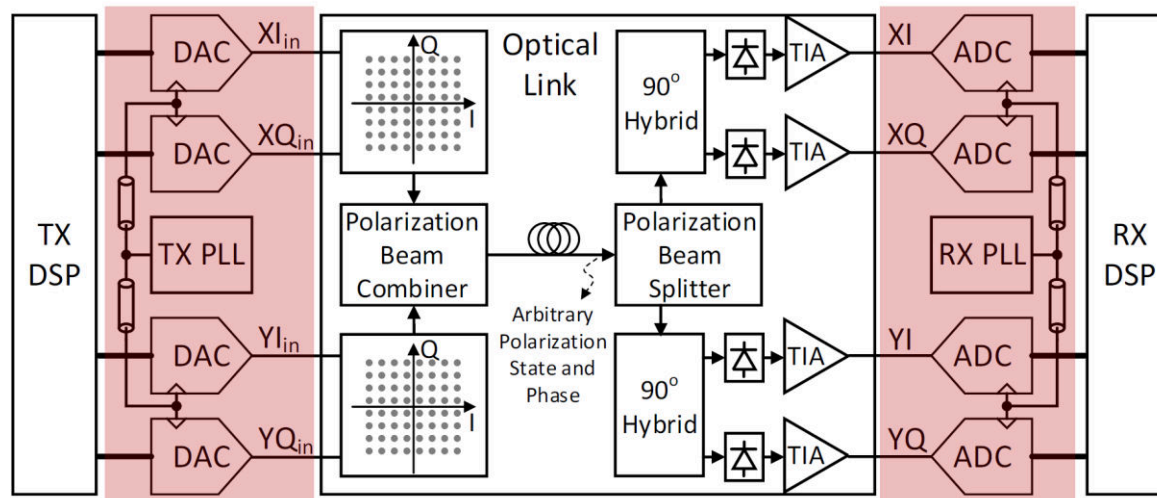


- State-of-the-art ADC/DCA brings revolutionary impact on communication systems

Application Example—High Speed Wireline



40-97GS/s DAC and ADC with 40GHz AFE Bandwidth (R. L. Nguyen, et. al., ISSCC 2021)



A 600 Gbps DP QAM 64 Coherent Optical Transceiver Frontend with 4 x 105 GSps 8 bit ADC/DAC in 16 nm CMOS (Guansheng Li, et. al., ISSCC 2024)

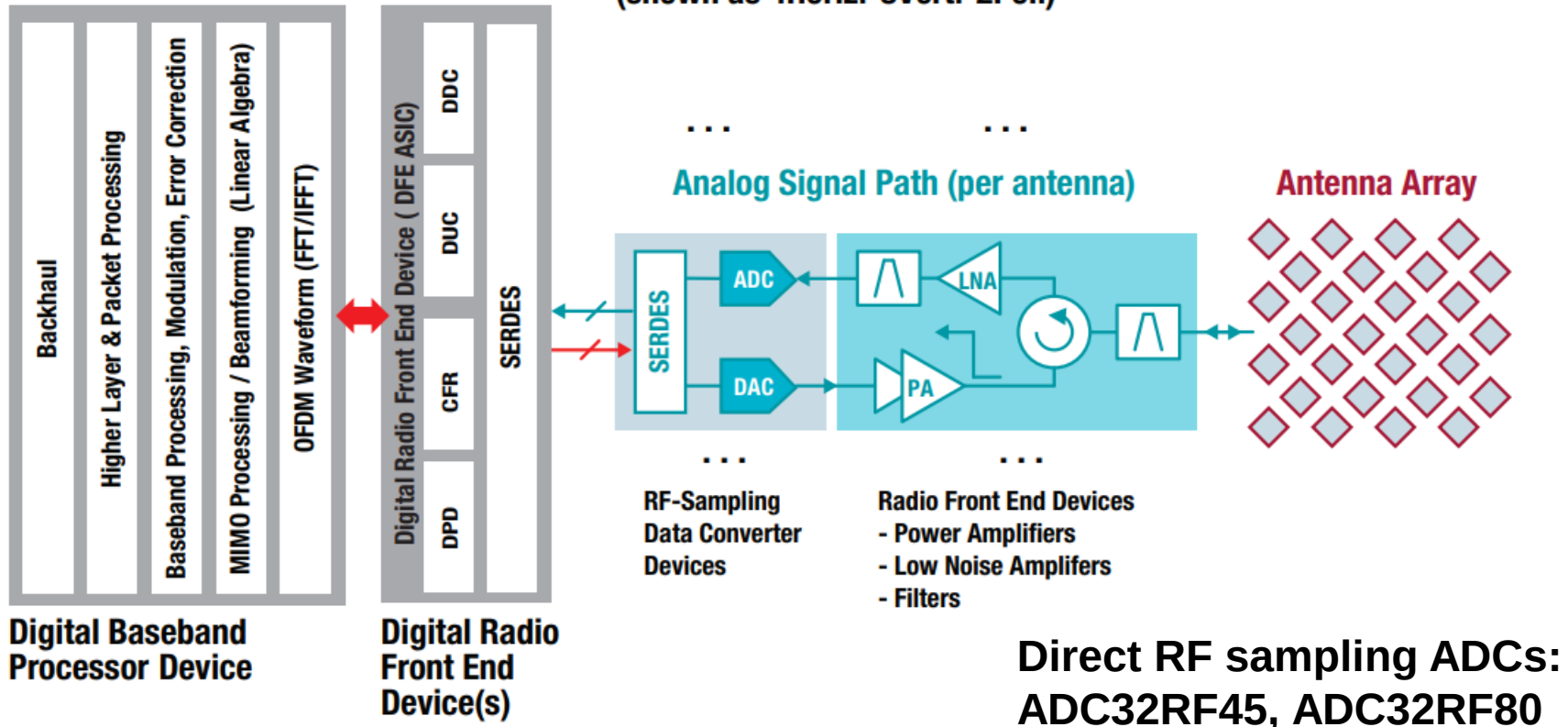


Standalone vs. Embedded

- Standalone ADCs/DACs
 - Very high performance (resolution, speed)
 - High power consumption/large chip size
 - Usually use the high performance analog processes
- Embedded ADCs/DACs
 - Has to use the standard CMOS technologies
 - Die area is limited
 - Usually modest performance (resolution, speed)

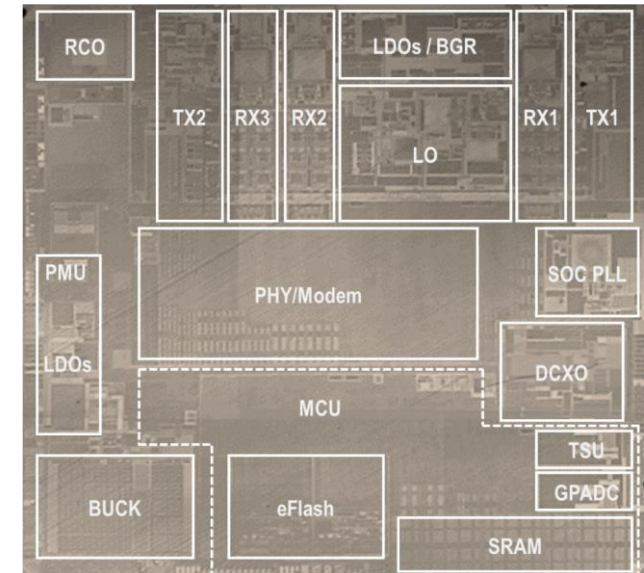
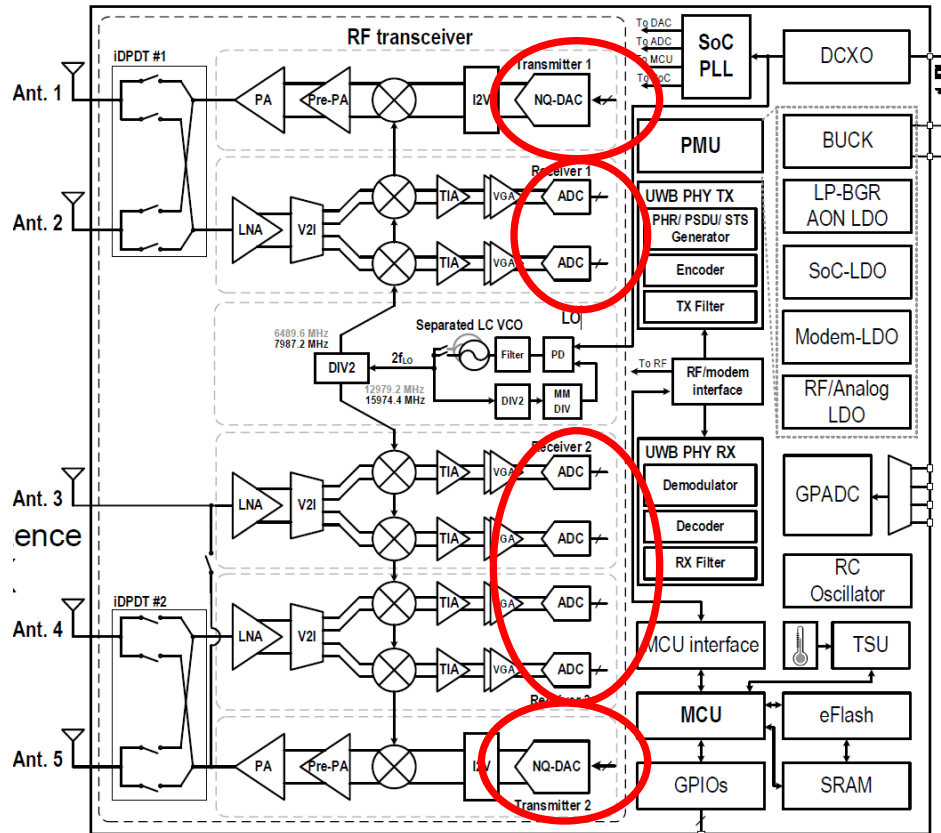
Application of Standalone ADC/DAC

Typical 64 Antenna MIMO Base Station (shown as 4Horiz.-8Vert.-2Pol.)



<http://www.ti.com/lit/wp/sszy025/sszy025.pdf> Analog advancements make waves in 5G wireless communications

Application of Embedded ADC/DAC



*Kim, et. al, ISSCC'23
(Samsung, 28nm CMOS)*

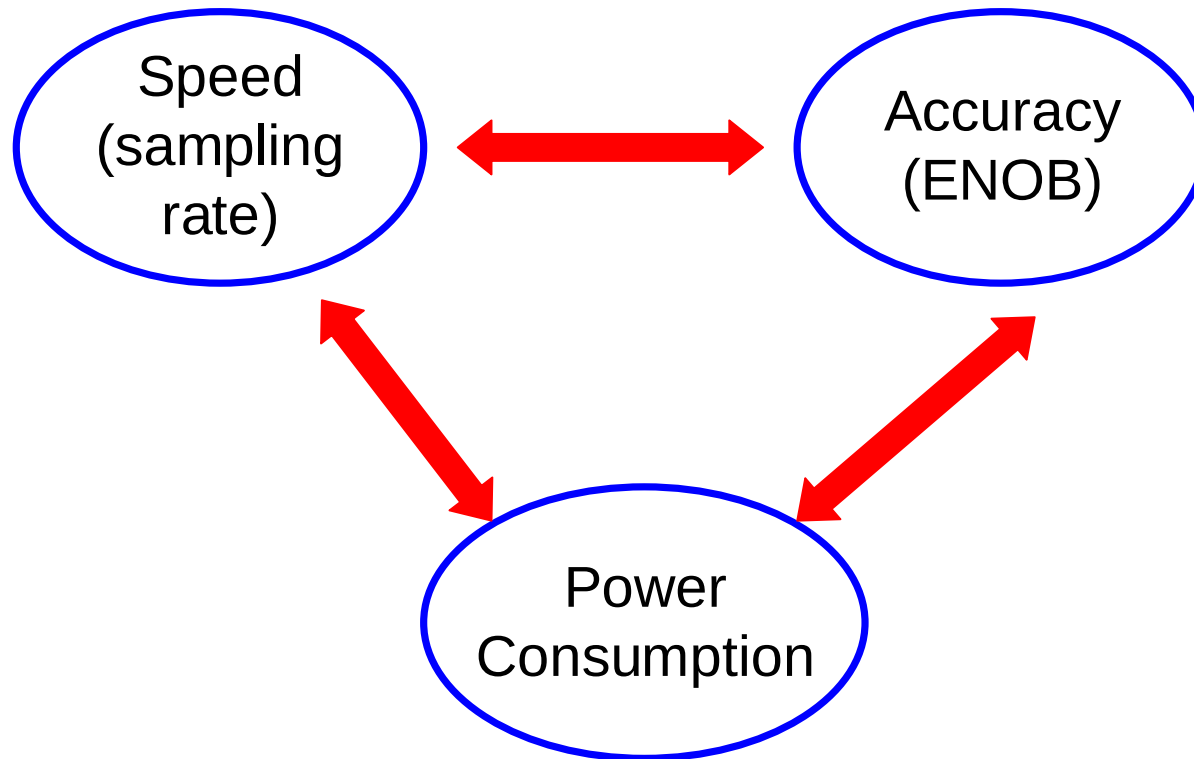
- ❑ Fully Integrated UWB SoC solution, supports IEEE 802.15.4/4z, integrates RF, MCU, Modem, PMU and CMU etc. in a Single-Chip
- ❑ Highly integrated SoC with RF/analog/digital in the same chip, requiring embedded ADC's/DAC's



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- **ADC/DAC Performance Spec.**
- ADC Architectures
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ADC/DAC Performance



Most commonly used
Figure of Merit (FoM)
(Unit: pJ/conv. step)

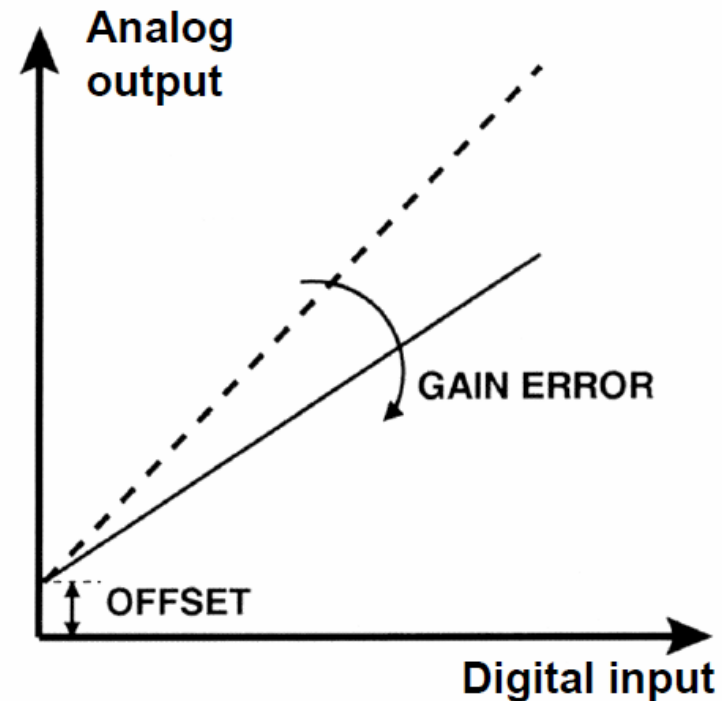
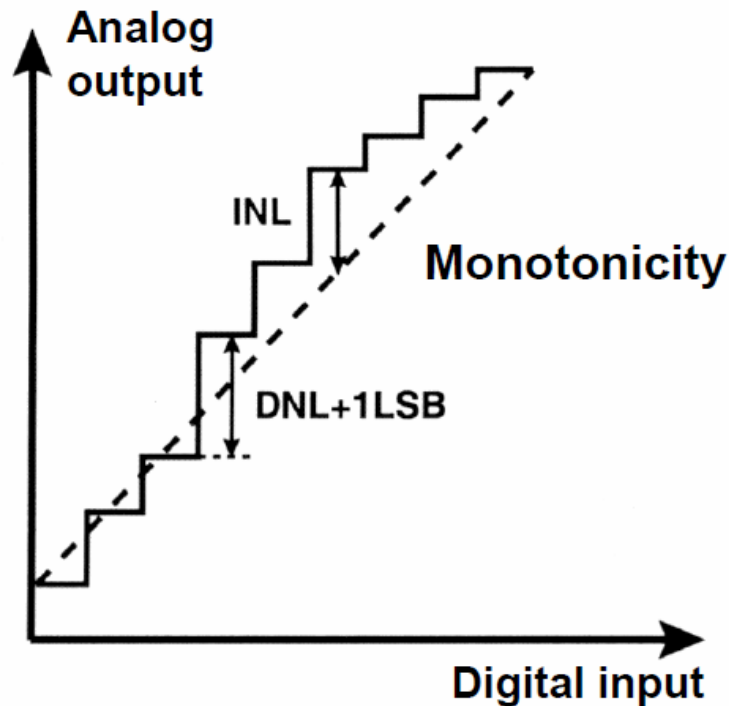
$$FoM = \frac{\text{Power Consumption}}{\text{Sampling Rate} \times 2^{ENOB}}$$



ADC/DAC Linearity Spec.

- Static Linearity
 - Gain error
 - Offset
 - Monotonicity
 - INL(integral nonlinearity)
 - DNL (differential nonlinearity)
- Dynamic linearity
 - Signal to noise ratio (SNR)
 - Spur free dynamic range (SFDR)
 - Signal to noise and distortion ratio (SNDR)
 - Effective number of bits (ENOB)
 - Performance drop for Nyquist sampling

Static Linearity

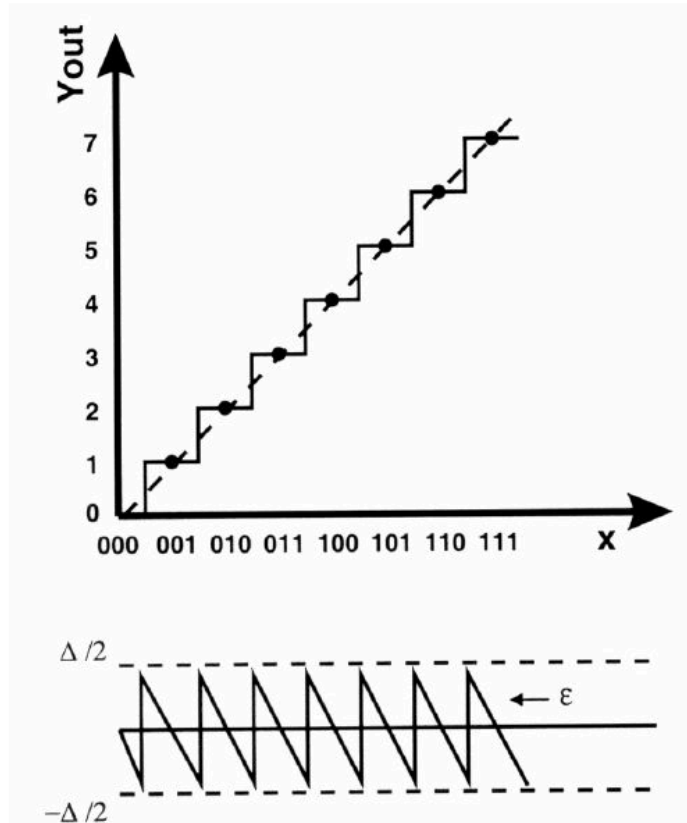


Differential Nonlinearity : $DNL = Y_{OUT}(B) - Y_{OUT}(B-1) - 1 \text{ LSB}$

Integral Nonlinearity : $INL = Y_{OUT}(B) - Y_{OUT,id}(B)$

W. Sansen, Analog Design Essentials

Quantization Noise and SNR



$$P_{\text{Noise}} = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} \epsilon^2 d\epsilon = \frac{\Delta^2}{12}$$

$$V_{\text{ptp}} = 2^N \Delta$$

$$P_{\text{Signal}} = \frac{V_{\text{ptp}}^2}{8}$$

$$\text{SNR} = \frac{3}{2} 2^{2N}$$

$$\text{SNR} = 6N + 1.76 \text{ dB}$$

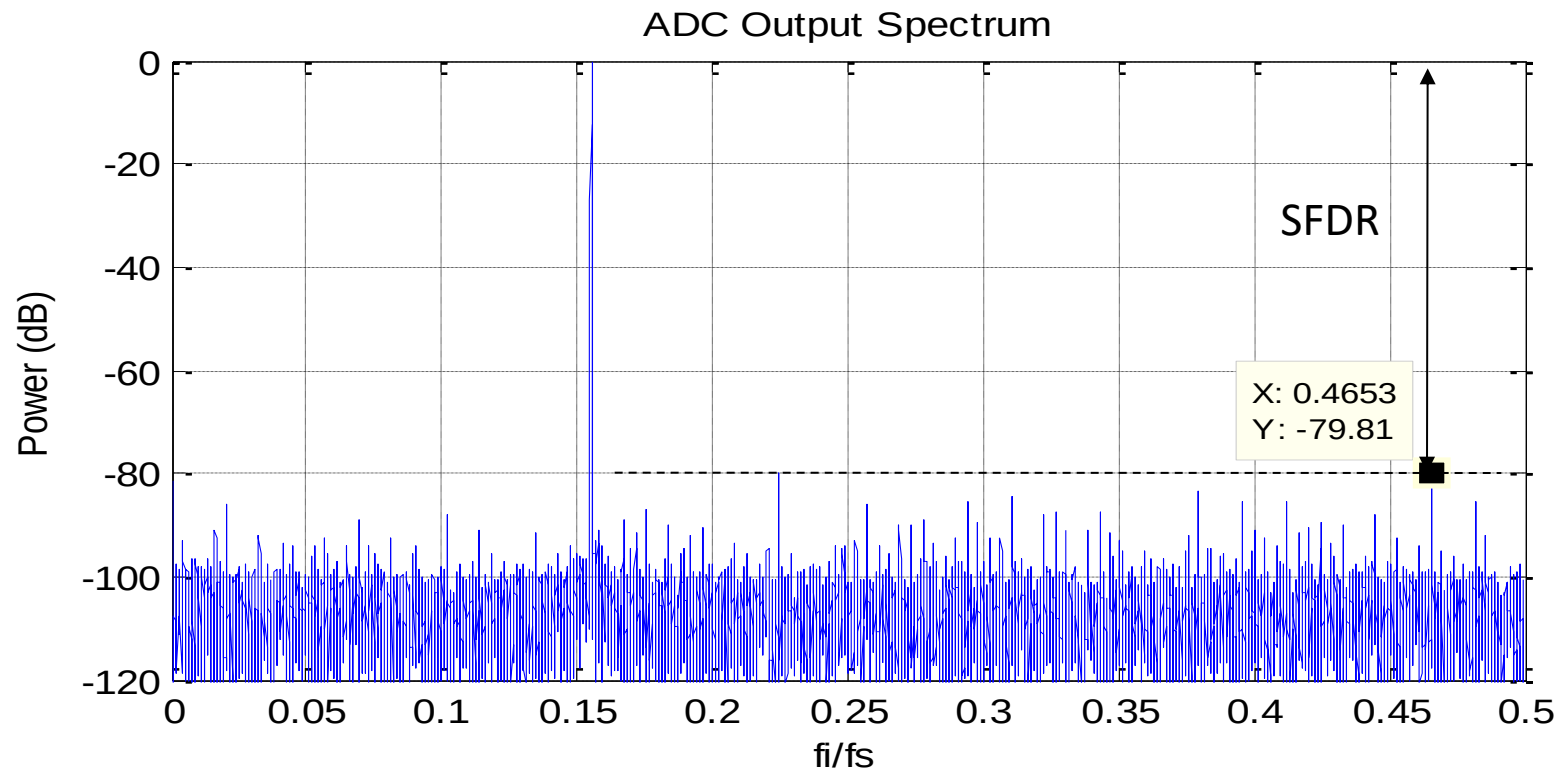
W. Sansen, *Analog Design Essentials*

Effective Number of Bits (ENOB)

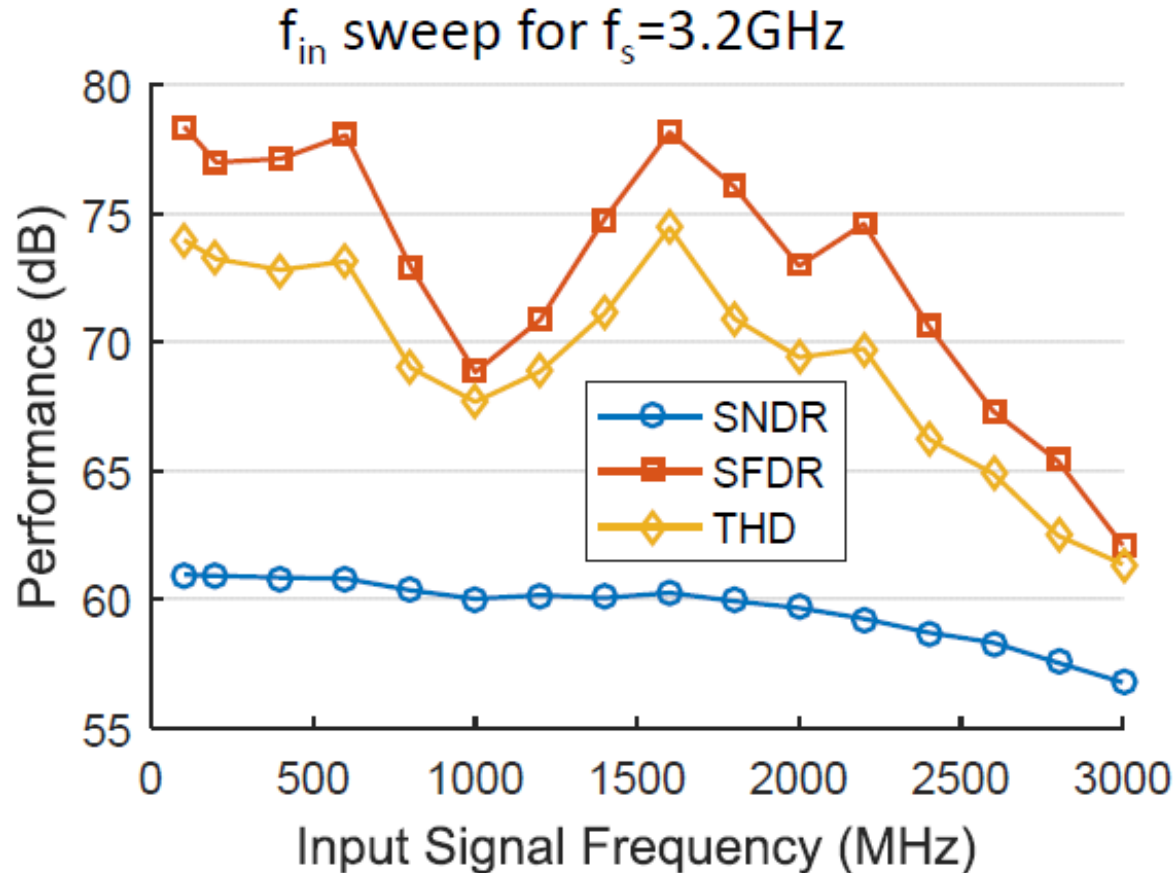
$$\text{ENOB} = \frac{\text{SNR(dB)} - 1.76}{6.02}$$



SFDR

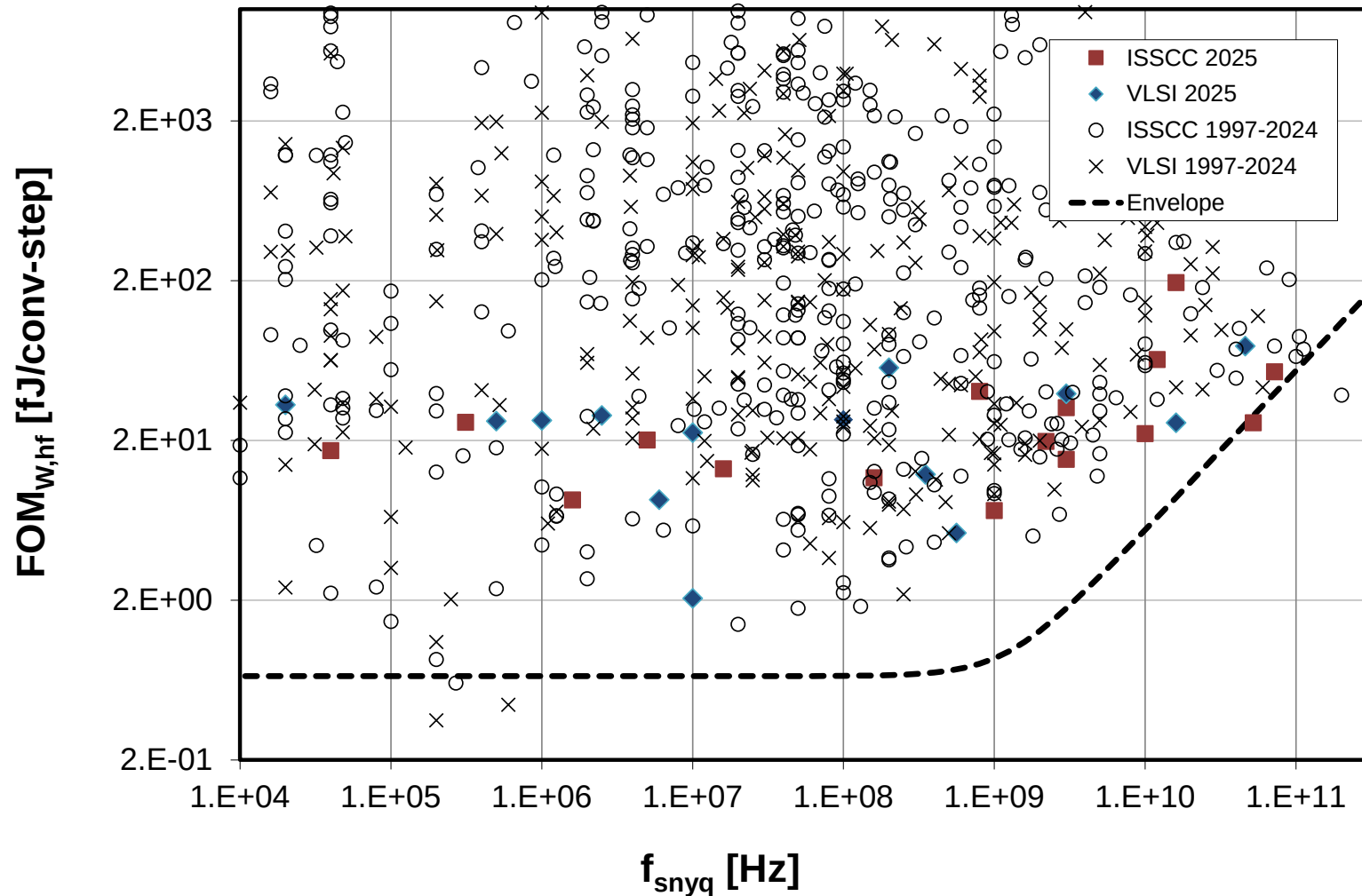


Performance Drop w.r.t. Input Frequency



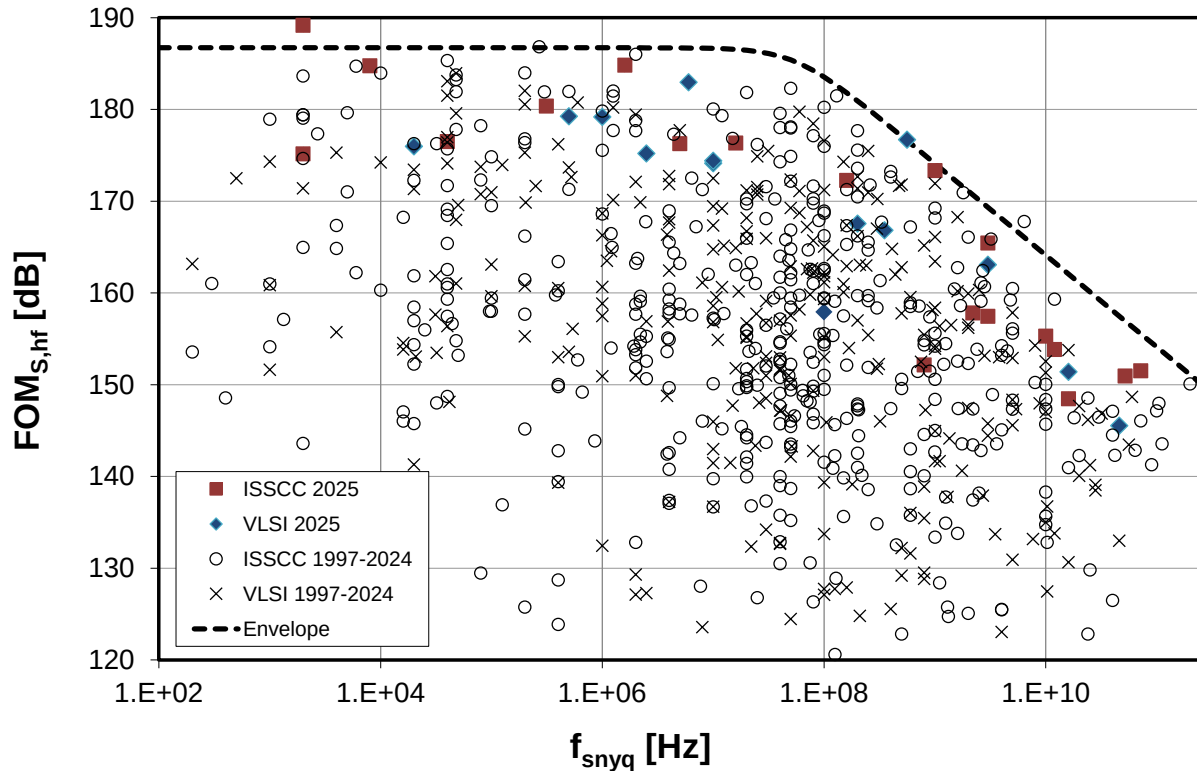
B. Hershberg, et.al., ISSCC'19

Walden FOM vs. Speed



ADC Performance Survey 1997-2025, by Boris Murmann

Schreier FOM vs. Speed

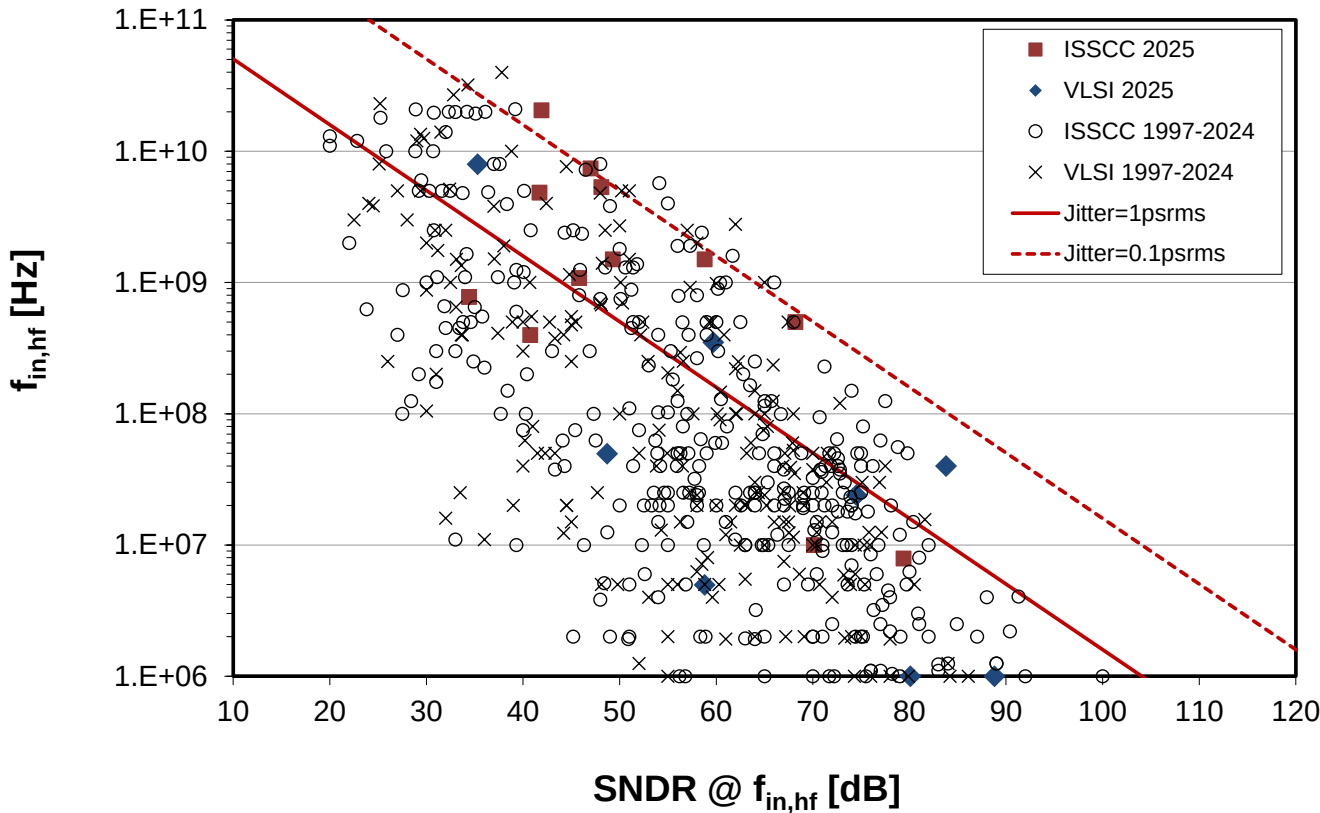


- ❑ At low sampling rates, limited by thermal noise, independent of sampling rate
- ❑ Higher speeds of operation present additional challenges for accuracy

Power normalized noise and distortion versus the Nyquist sampling rate.

ADC Performance Survey 1997-2025, by Boris Murmann

Aperture



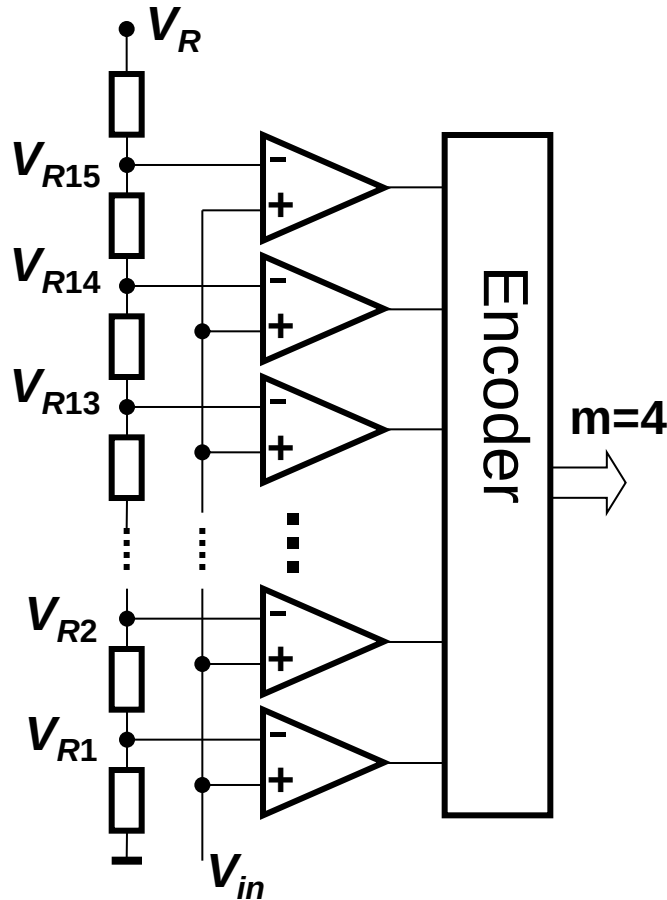
- ❑ Sampling jitter or aperture errors make achieving both high resolution and high bandwidth particularly difficult
- ❑ 1ps jitter $\rightarrow$ 0.1ps jitter in the past 10 years



Outline

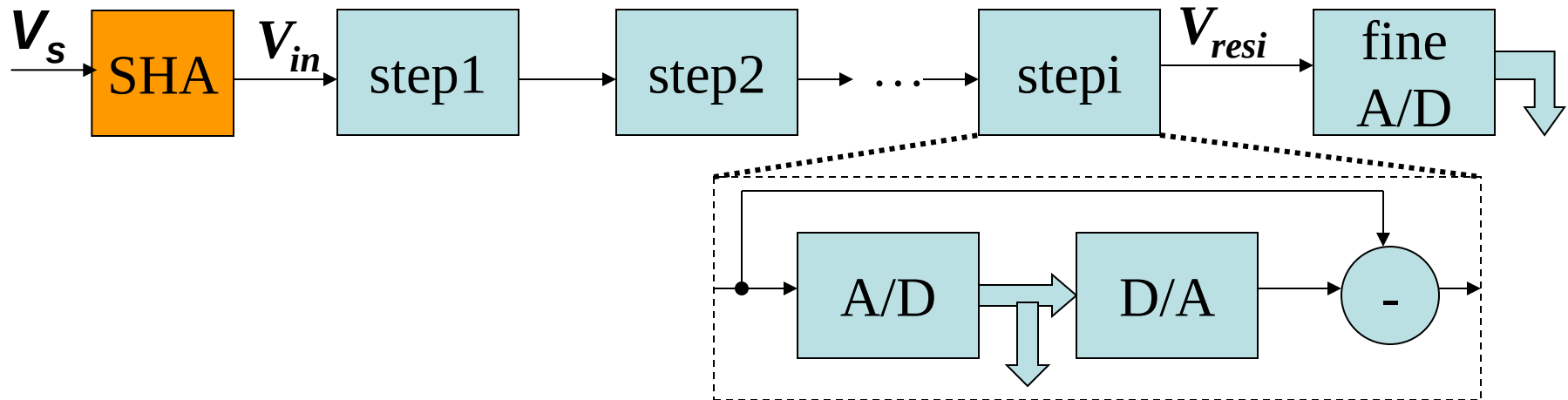
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Nyquist ADC Architectures – Flash ADC



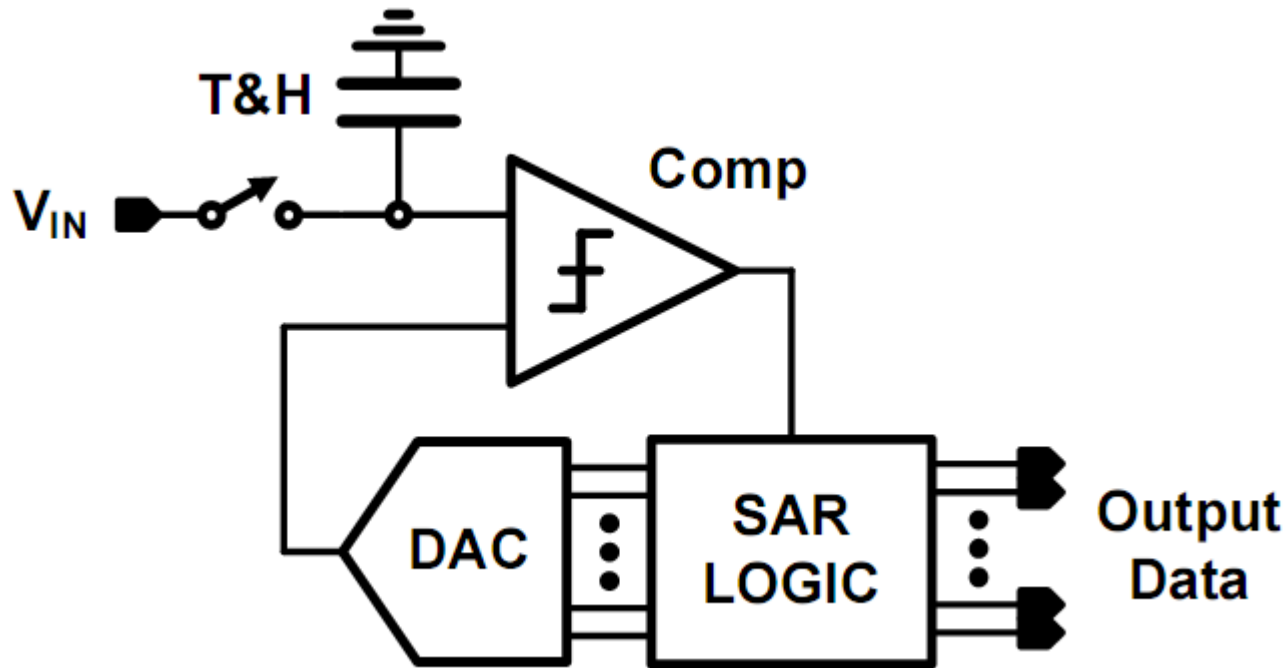
- ❑ High speed
- ❑ High power consumption
- ❑ Low resolution
- ❑ High complexity

Nyquist ADC Architectures – Pipelined ADC



- ❑ Modest high speed
- ❑ Modest high resolution
- ❑ High power consumption
- ❑ High circuit complexity

Nyquist ADC Architectures – SAR ADC

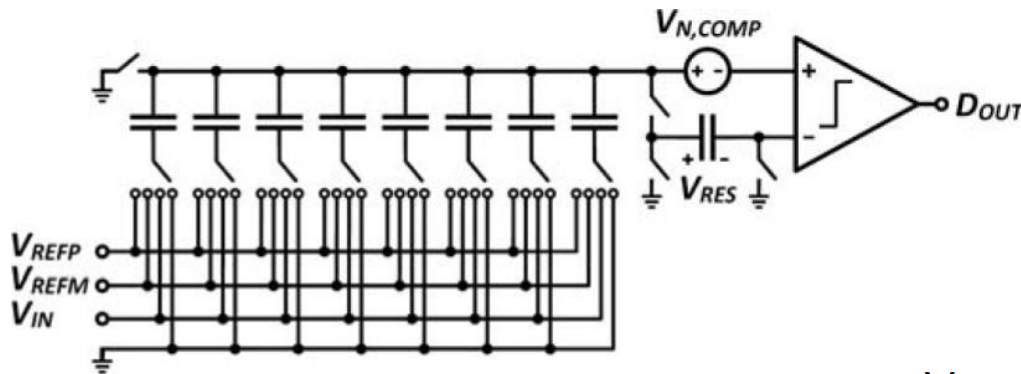


Very high speed by interleaving

- ☐ Modest high speed
- ☐ Low to modest high resolution
- ☐ Low power consumption
- ☐ Low circuit complexity

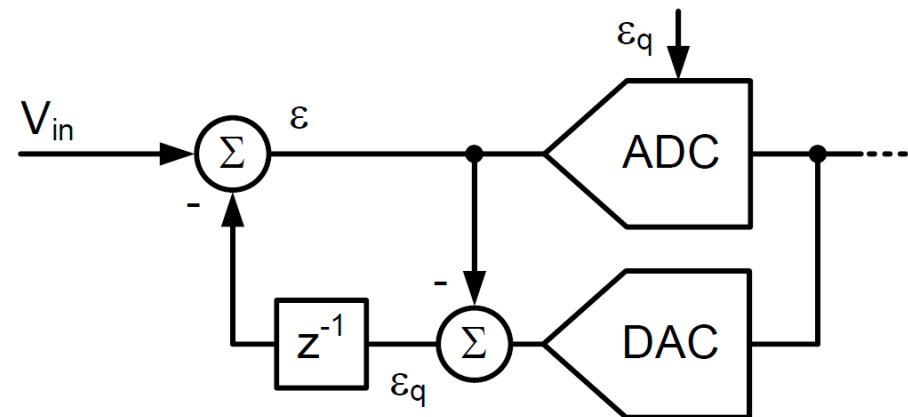
Noise Shaping ADC

- 1st noise shaping ADC at ISSCC'12 (Fredenburg & Flynn)
- The residue voltage is sampled on another resistor and feedback to next run of SAR operation
- The input-referred noise of the comparator is noise-shaped along with the quantization noise

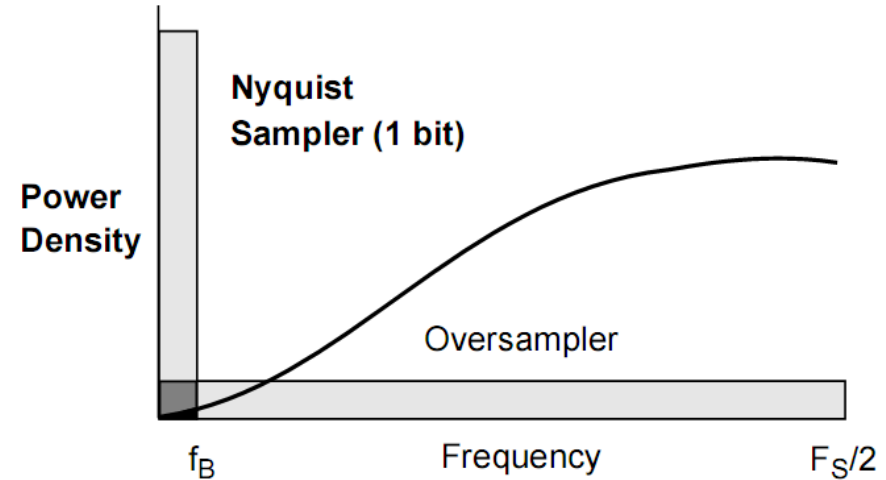
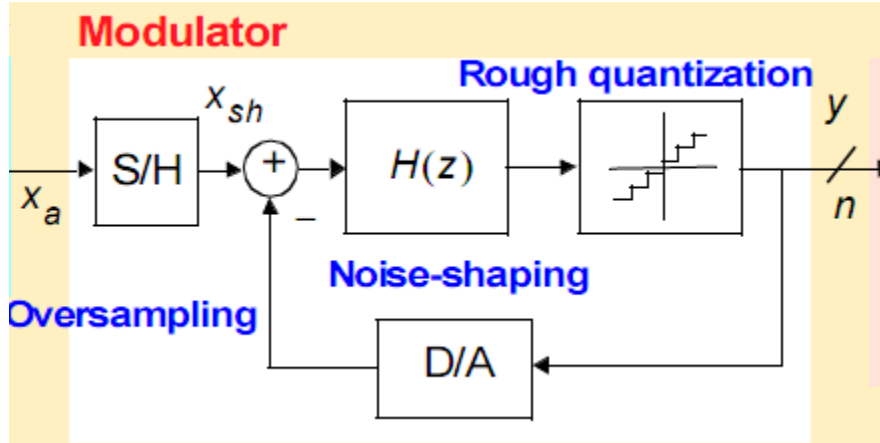


$$D_{OUT}(z) = -V_{IN}(z) + \frac{1}{1+z^{-1}}[Q(z) + V_{N,COMP}(z)]$$

J. Fredenburg & M. Flynn, "A 90MS/s
11MHz bandwidth 62dB SNDR noise-
shaping SAR ADC," Digest of ISSCC'12



Oversampled ADC – $\Sigma\Delta$ ADC



- ✓ Oversampling
- ✓ Noise shaping
- ✓ Low-pass filtering

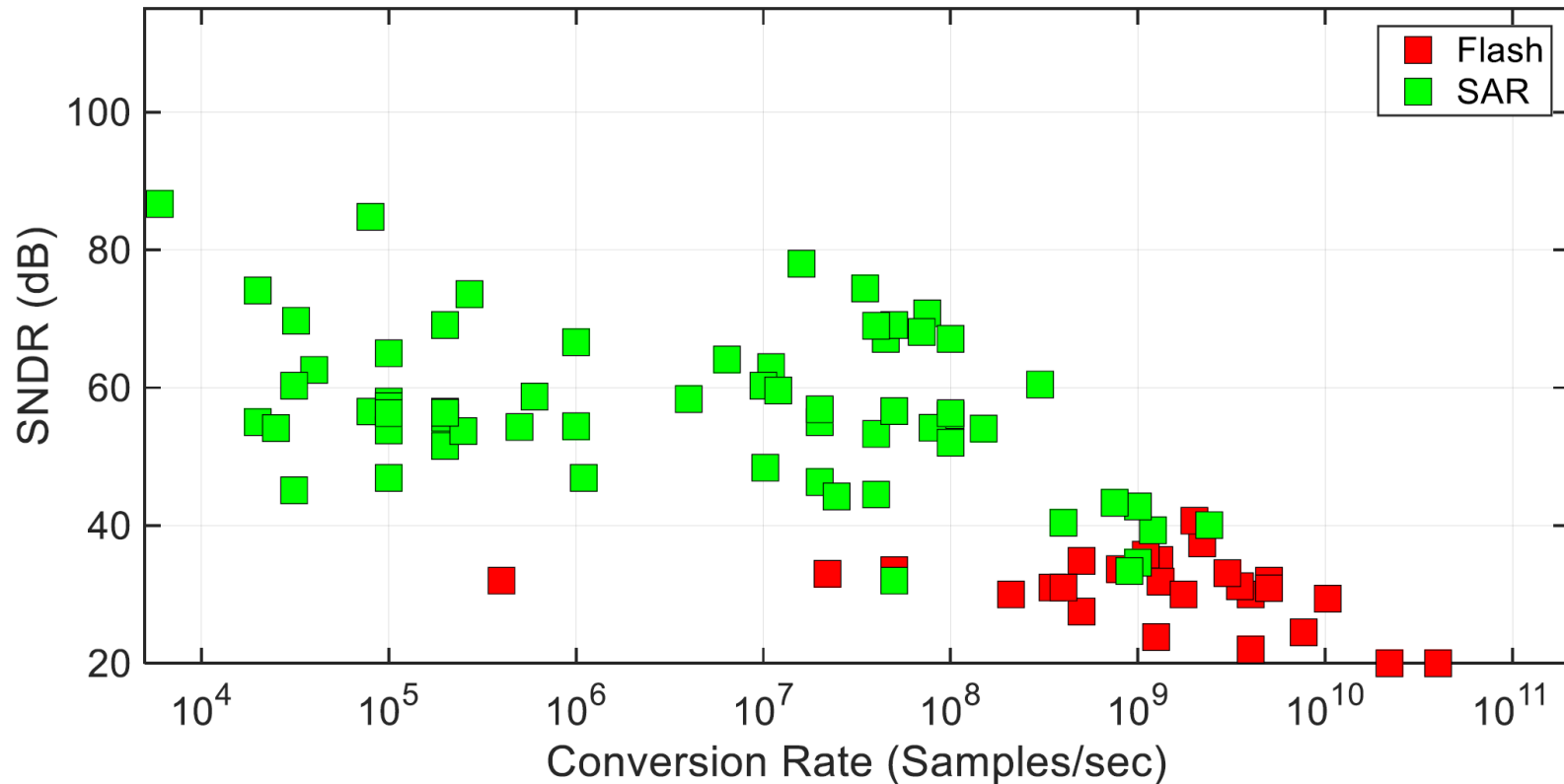
- ❑ Low speed
- ❑ High resolution
- ❑ Low power consumption



Other ADC Architecture

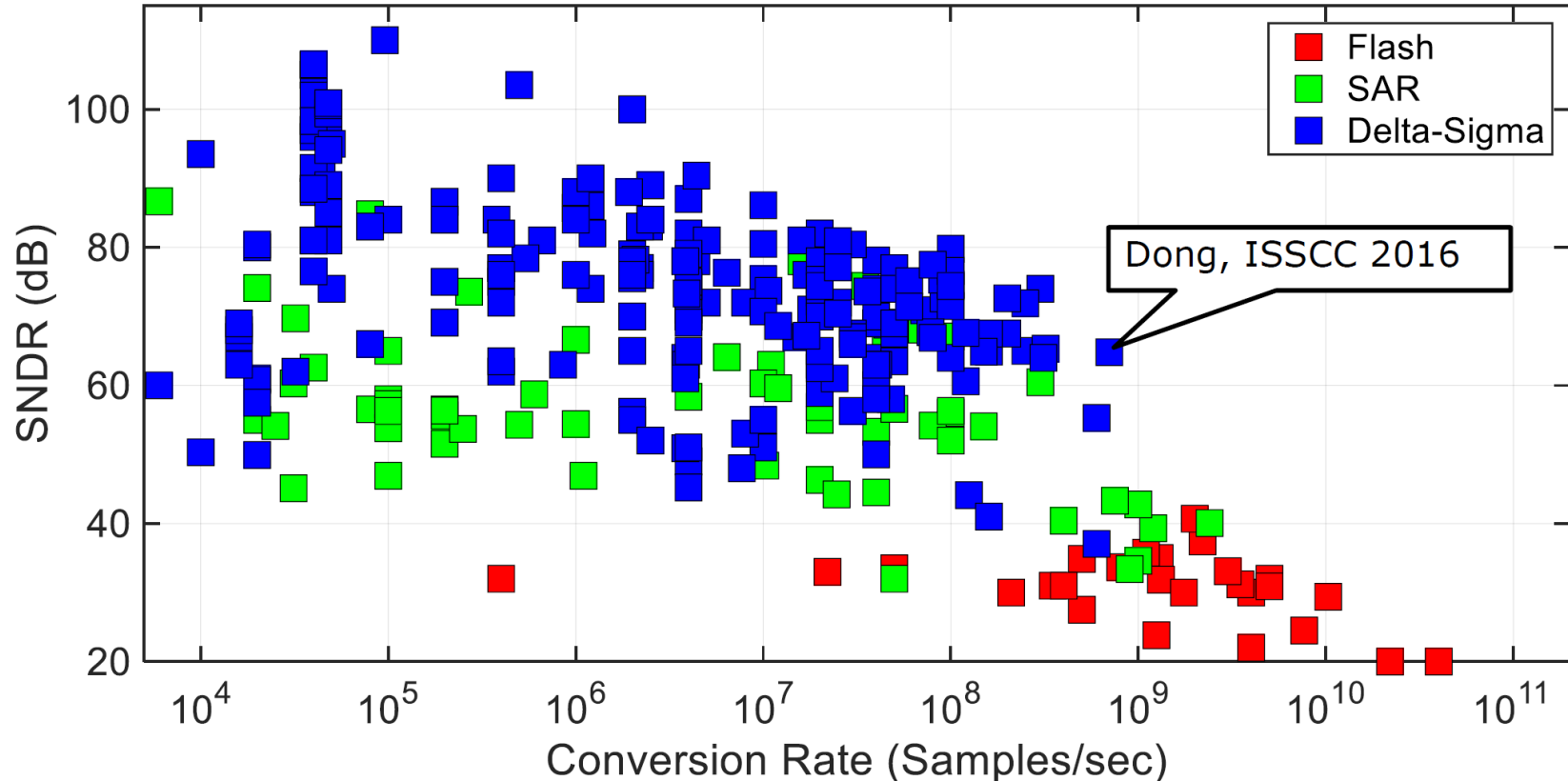
- Dual-slope ADC (voltage-time-digital)
- VCO based ADC (voltage-frequency/phase-digital)
- Cyclic ADC (single pipelined stage + cyclic operation)

Comparison of Different Types of ADCs



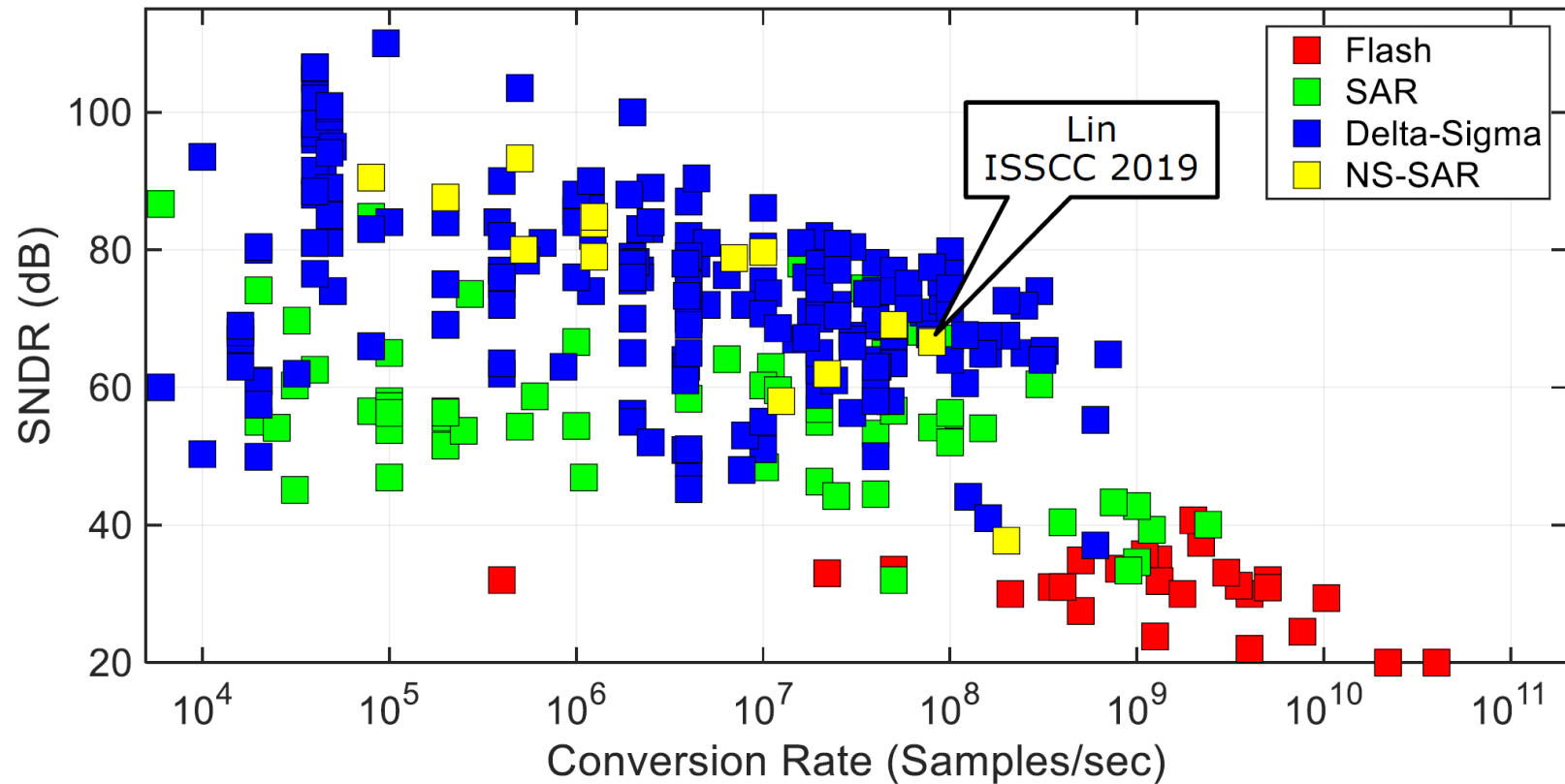
Courtesy: Boris Murmann, Introduction to ADCs/DACs

Comparison of Different Types of ADCs



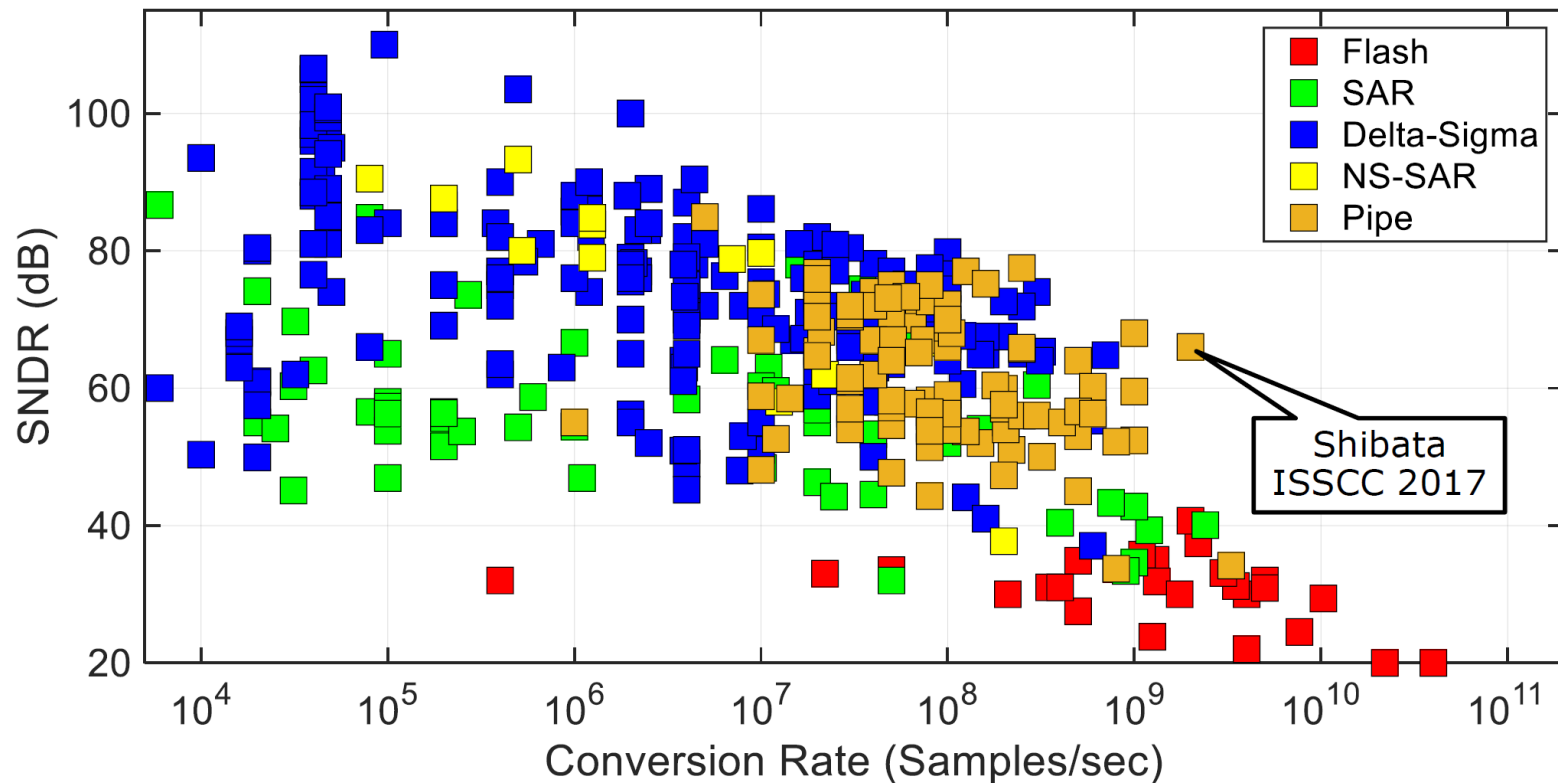
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Comparison of Different Types of ADCs



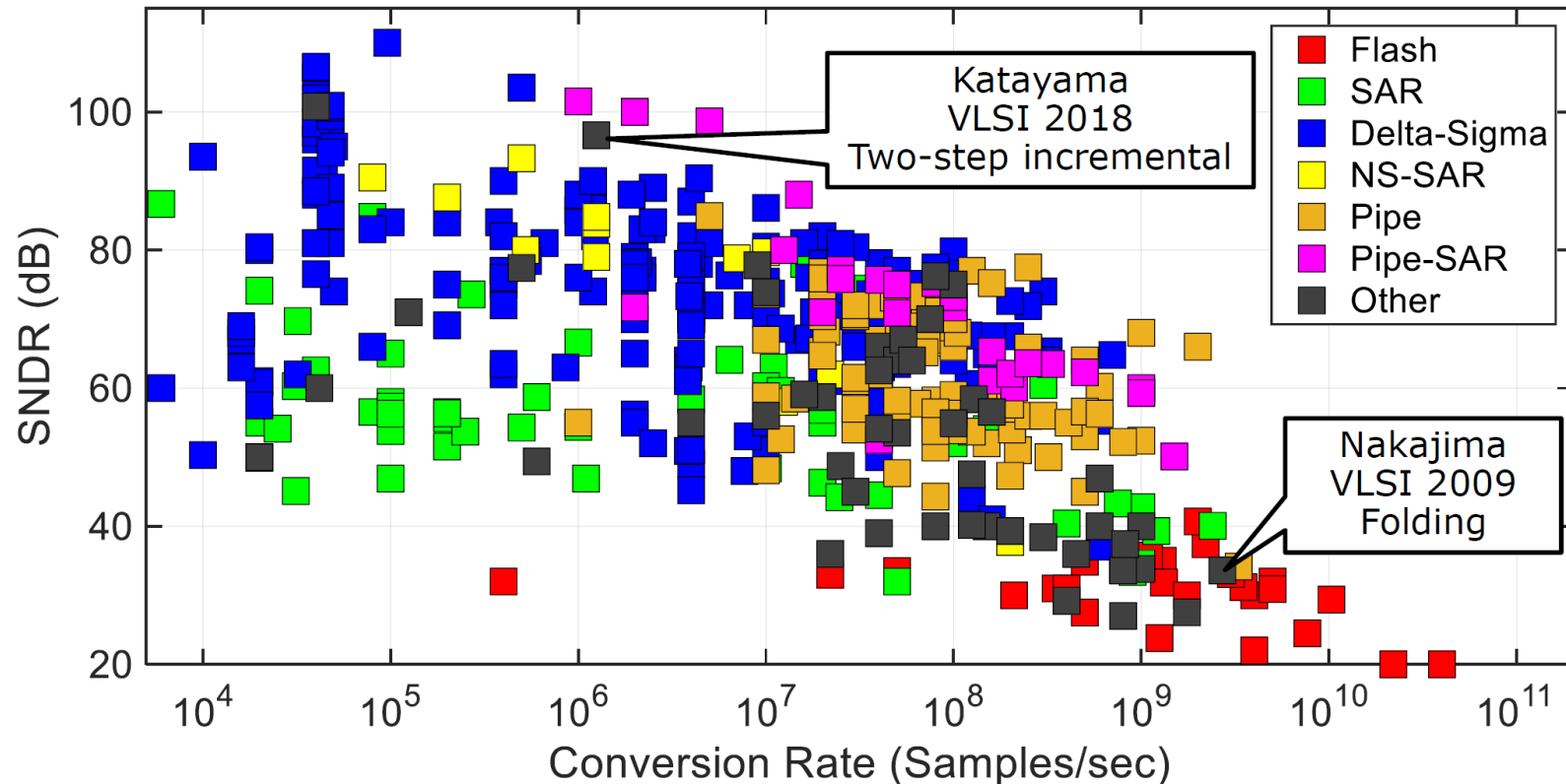
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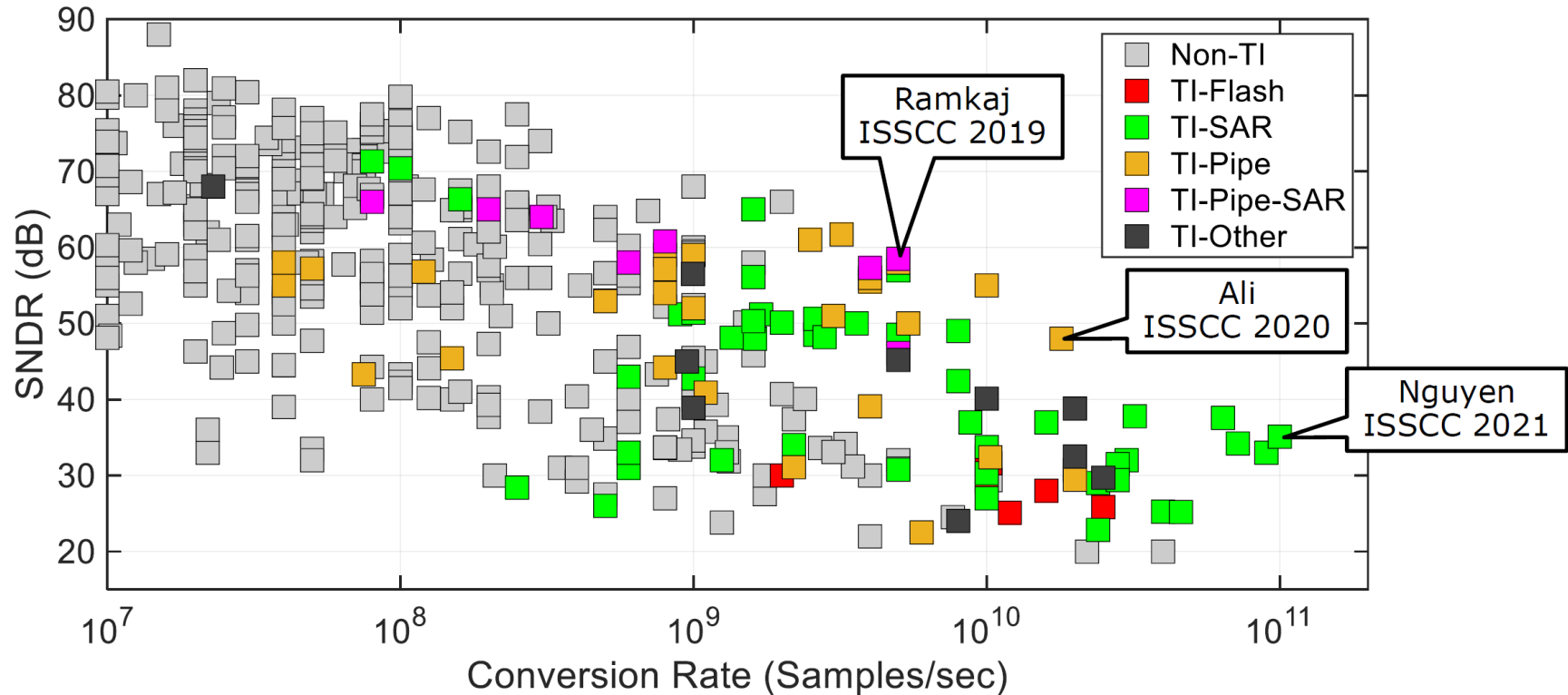
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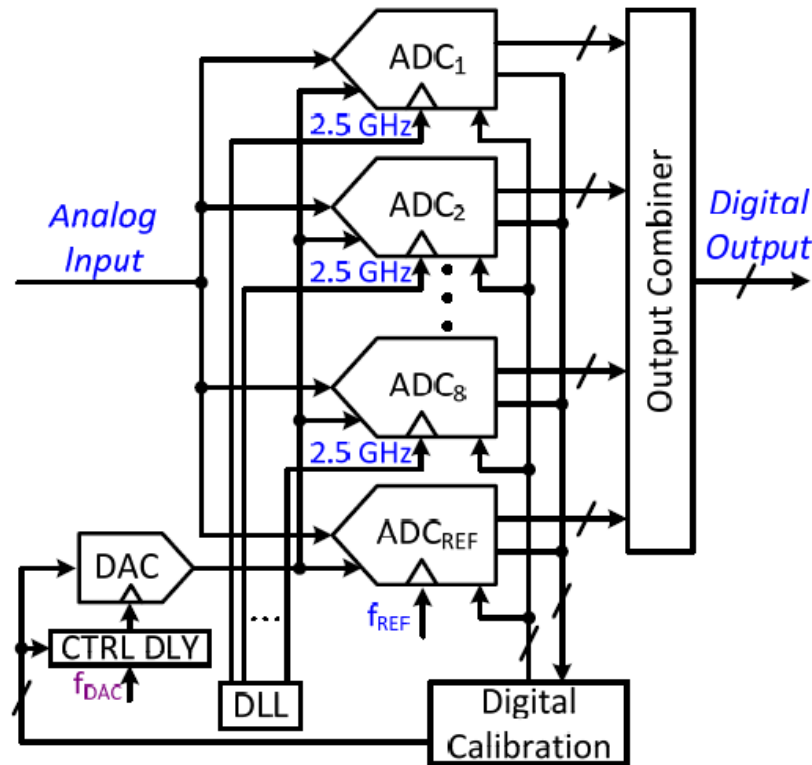


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High speed ADC – Flash ADC

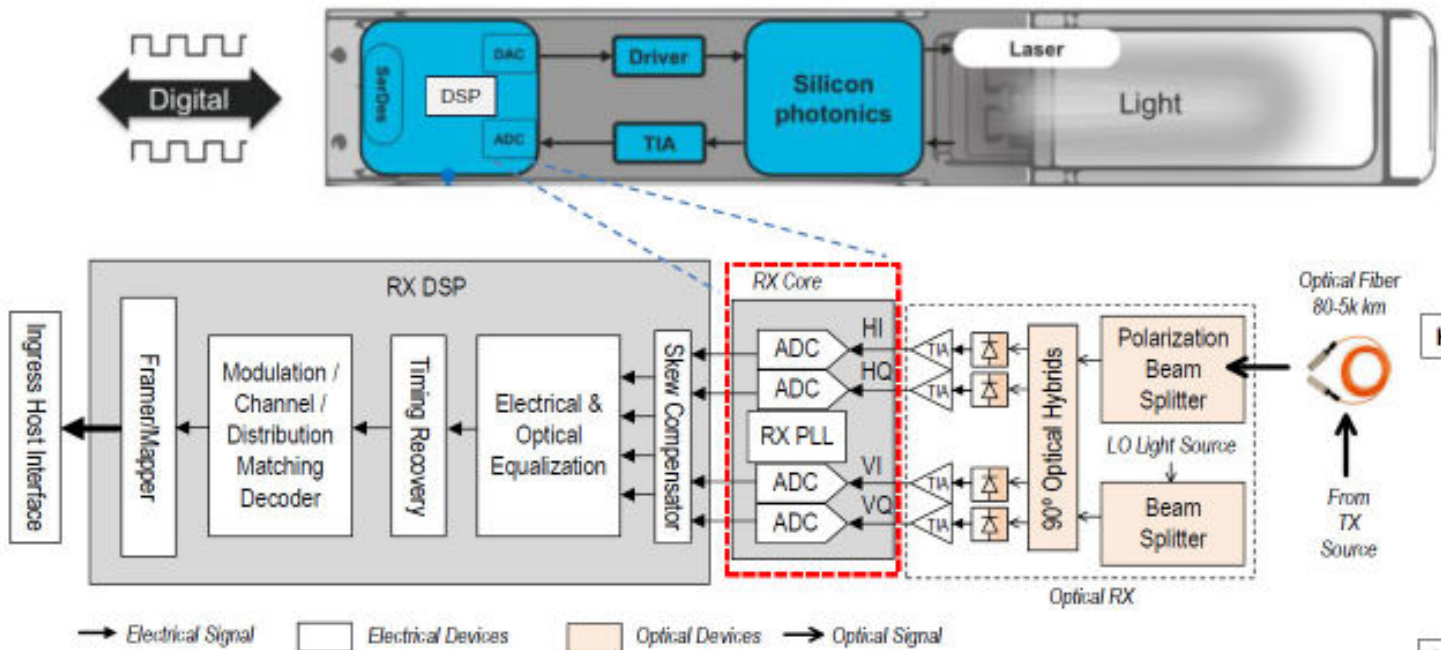
- Sub-ADC: 6bit
2.5Gsp/s Flash ADC
- 8x interleaving
- 124 fJ/conv-step



Highest? No!

Chen & Pileggi, "A 69.5mW 20GS/s 6b Time-Interleaved ADC with Embedded Time-to-Digital Calibration in 32nm CMOS SOI" ISSCC'14

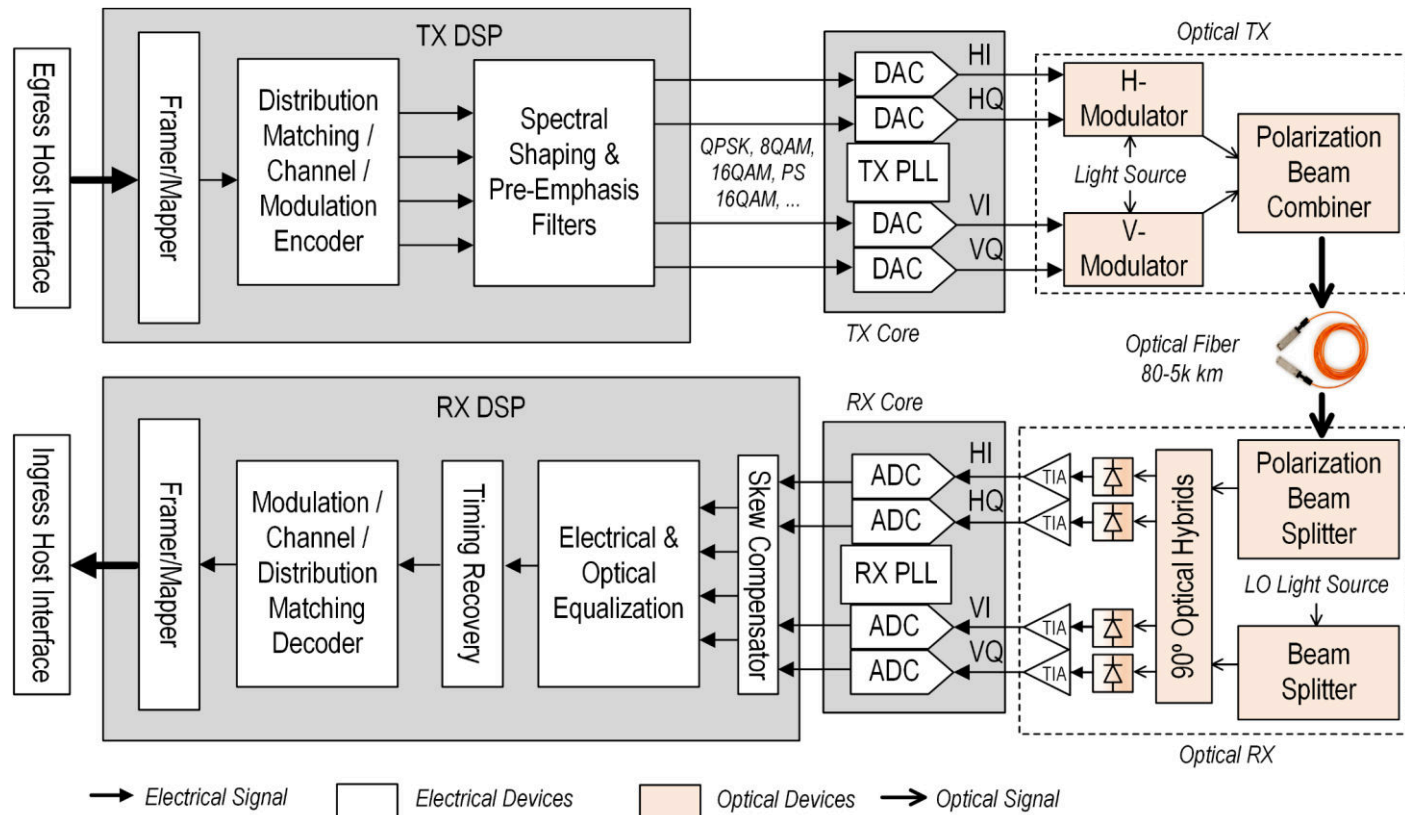
ADC with Highest Clock Frequency Published in ISSCC So Far



Nguyen, et. al., A 200 GS/s 8 b 20 fJ/c s Receiver with > 60 GHz AFE Bandwidth and 256 way Interleaved SAR ADC for 800 Gbps Coherent Communications in 5 nm FinFET, ISSCC'24 ([Marvell](#))

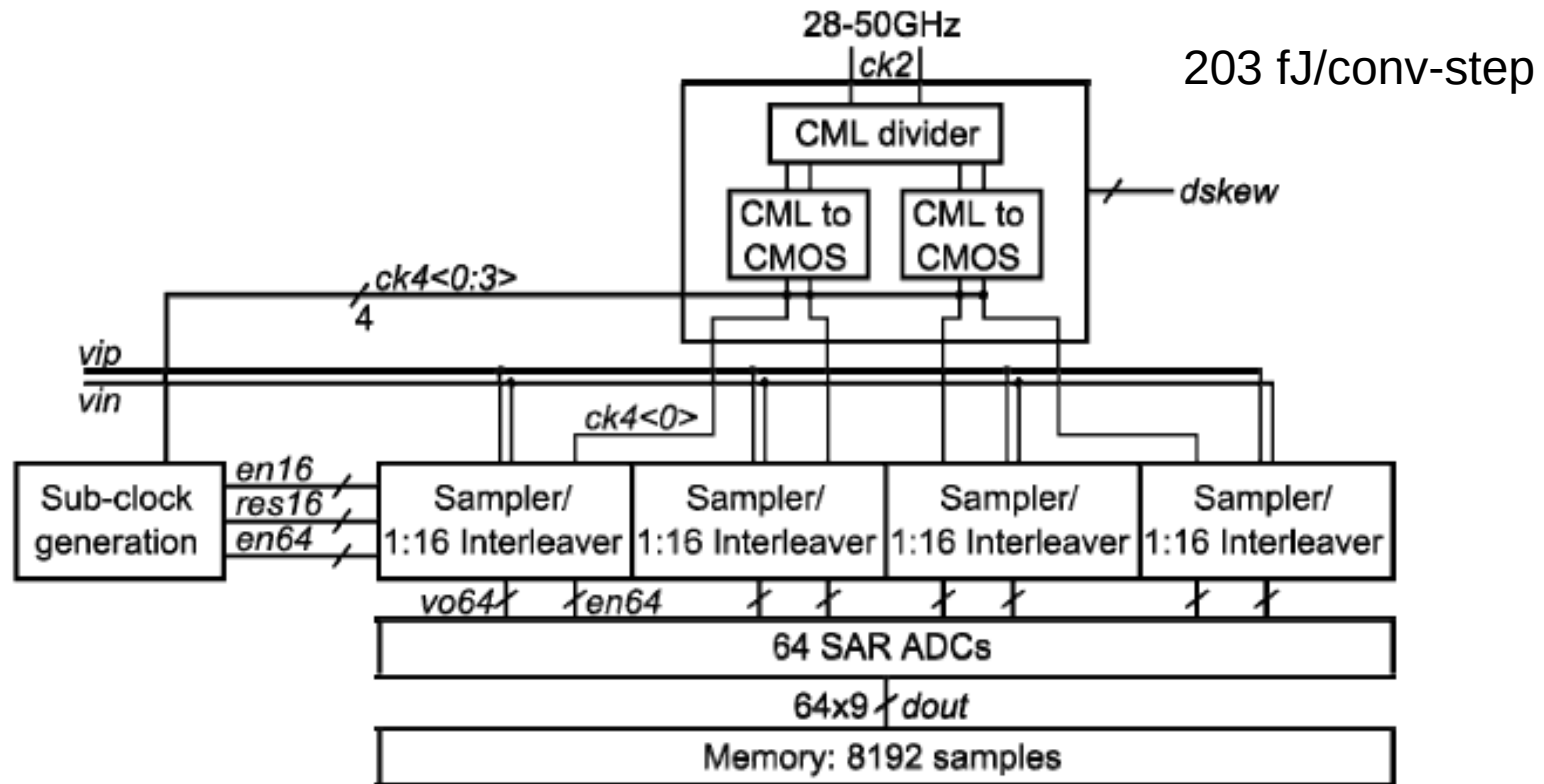
ADC with Highest Clock Frequency

Published in ISSCC So Far



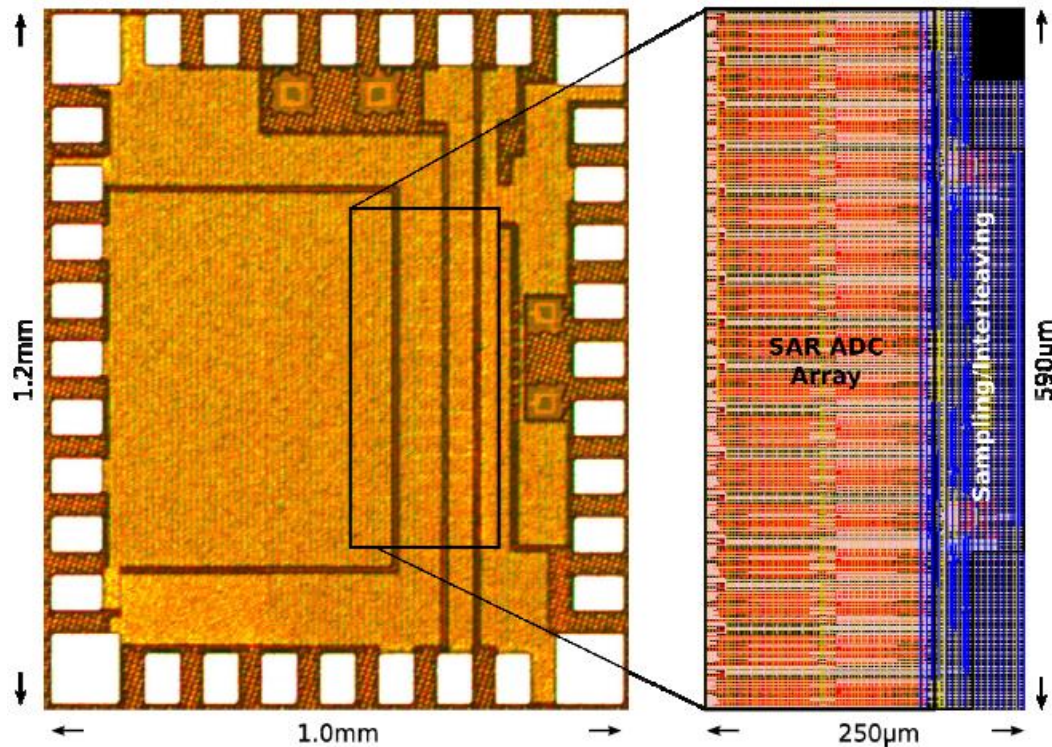
Nguyen, et. al., A Highly Reconfigurable 40-97GS/s DAC and ADC with 40GHz AFE Bandwidth and Sub-35fJ/conv-step for 400Gb/s Coherent Optical Applications in 7nm FinFET, **ISSCC'21** (Inphi)

ADC with Highest Clock Frequency Published in ISSCC So Far

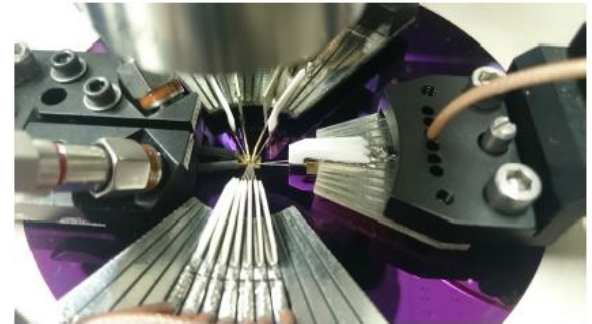


Kull, et. al., A 90GS/s 8b 667mW 64x Interleaved SAR ADC in 32nm Digital SOI CMOS, **ISSCC'14** (IBM)

ADC with Highest Input Frequency Published in ISSCC So Far



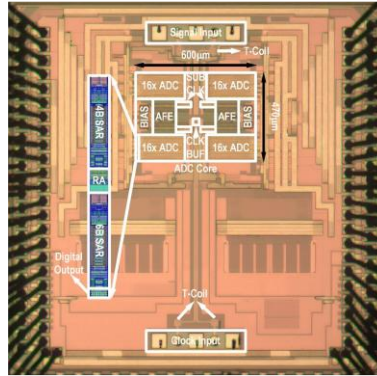
- Input frequency: 36GHz –the first with >30GHz input
- Only 0.15mm²



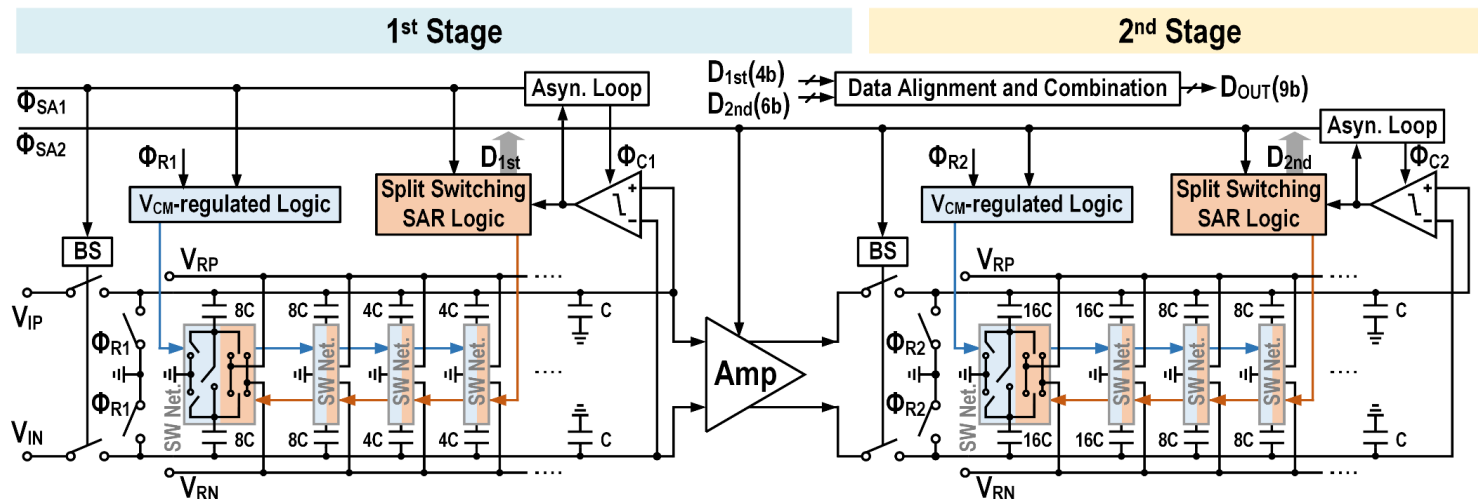
Kull, et. al., A 24-to-72GS/s 8b Time-Interleaved SAR ADC with 2.0-to- 3.3pJ/conversion and >30dB SNDR at Nyquist in 14nm CMOS FinFET, **ISSCC'18** (IBM)

ADC with Highest Input Frequency

Published in ISSCC So Far*



- 64-ch interleaving
- 1.125Gsps 2-Stage Pipeline SAR ADC
- Core area 0.28mm²

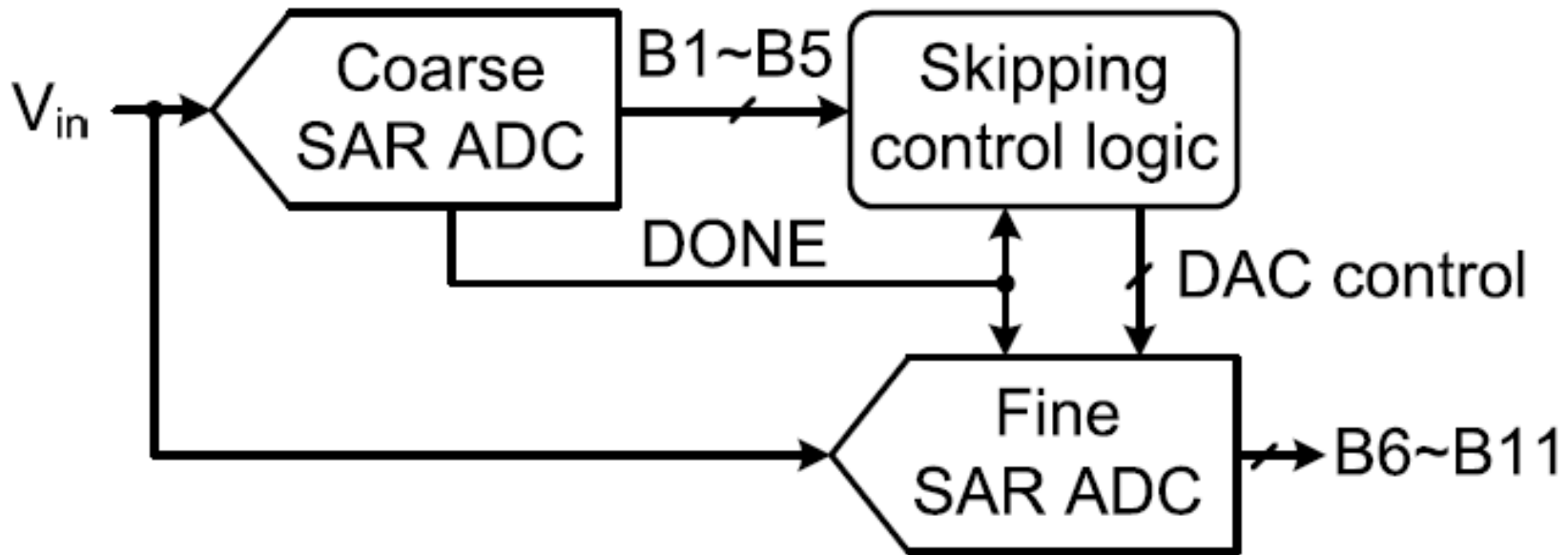


Zhang, et. al., A 72GS/s 9b Time Interleaved Pipeline SAR ADC Achieving 55.3/49.3dB SFDR at 20GHz/Nyquist Inputs in 16nm FinFET, **ISSCC'25** (U. Macau)

- Flash ADC is all-analog → benefit from process advancement is limited
- SAR ADC is more “digital” → scalable with process → more interleaving with the newest process
- SAR ADC is replacing Flash ADC for high speed applications



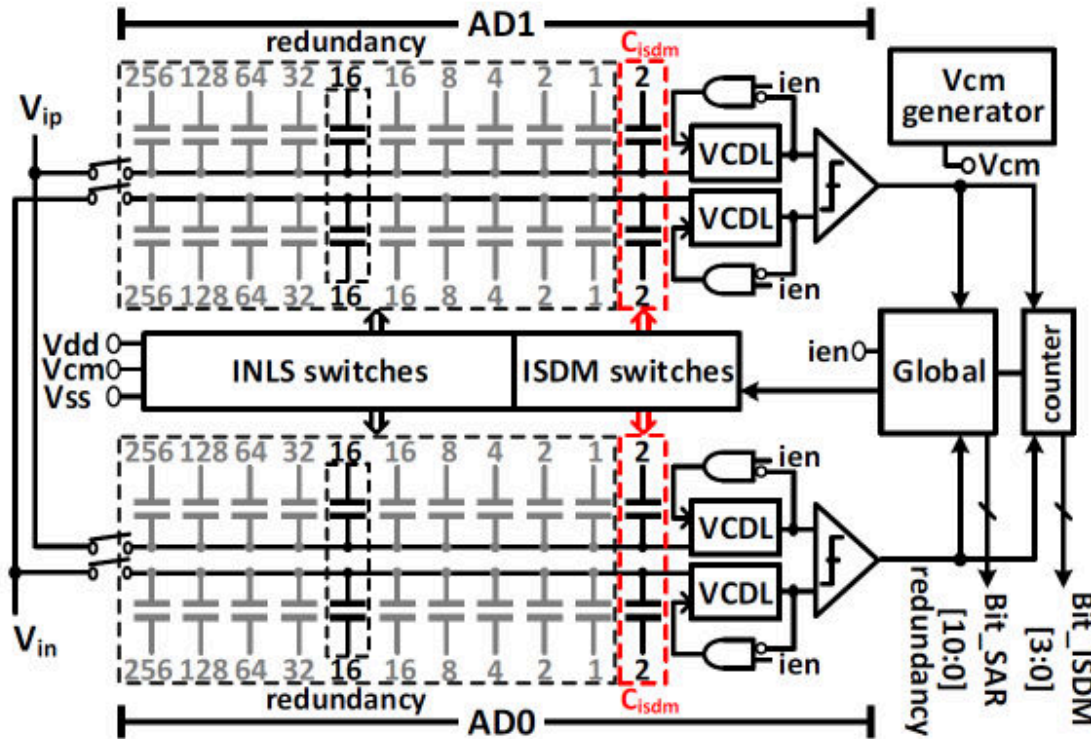
SAR ADC is ALSO favor of high energy efficiency



Tai, et. al., A 0.85fJ/conversion-step 10b 200kS/s Subranging SAR ADC in 40nm CMOS, ISSCC'14

- ❑ Coarse ADC to reduce # of switching of Fine ADC, hence to save power
- ❑ ENOB ~9 bit
- ❑ 0.0065mm² active area in 40nm CMOS
- ❑ 84nW @ 0.45V

SAR ADC is ALSO favor of high energy efficiency

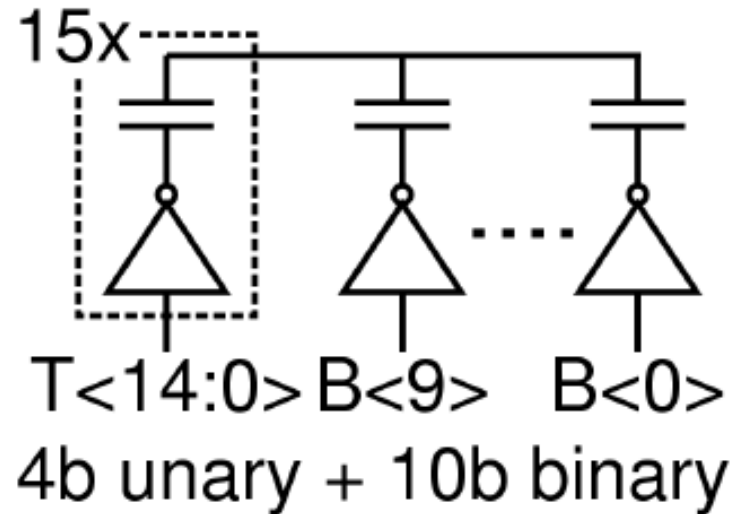


- **11b SAR ADC**
 - Two sub 9b CDAC
 - AD1: $0 \sim +2V_{dd}$ AD0: $0 \sim -2V_{dd}$
 - x2 swing
- **4b ISDM**
 - 5.4fF unit C
- **Redundancy**
 - Inter stage: 2b
 - $\pm 64\text{LSB}$ redundancy for $\pm 21\text{LSB}$ VCDL offset

Hsieh, et. al., A 0.4V 13-bit 270KS/s SAR-ISDM ADC with Opamp-Less Time-Domain Integrator, ISSCC'18

0.61fJ/conversion-step -- Lowest so far

High resolution SAR ADC

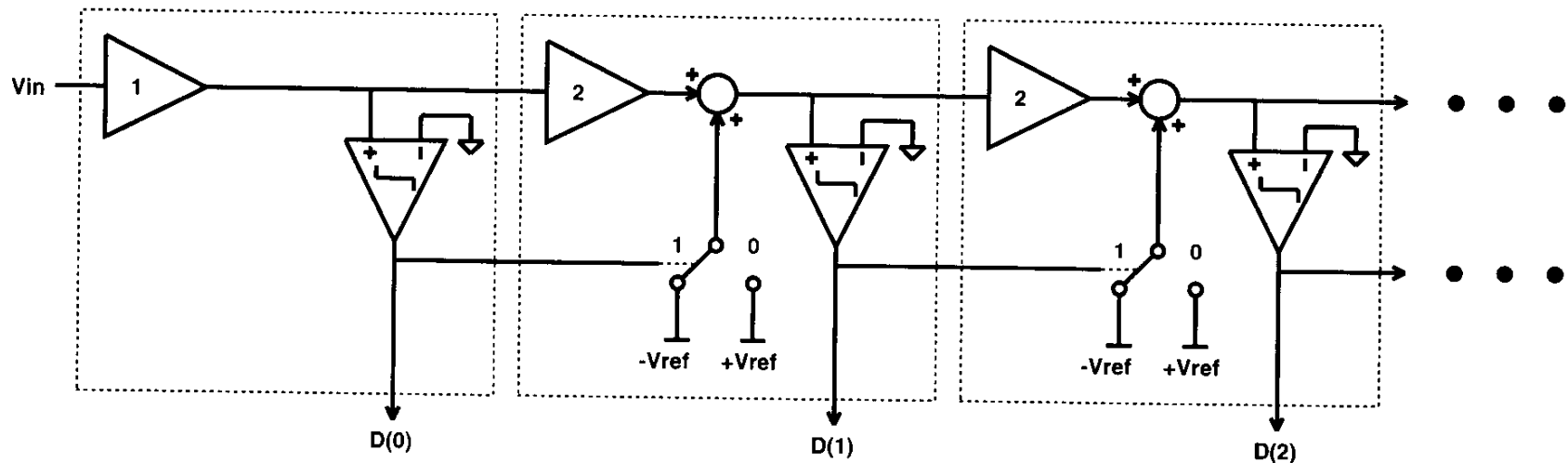


Harpe, et. al., An Oversampled 12/14b SAR ADC with Noise Reduction and Linearity Enhancements Achieving up to 79.1dB SNDR , ISSCC'14

- ❑ Oversampling to lower quantization noise floor
- ❑ Data driven noise reduction
- ❑ <1.4uW 8.2fJ/conv.-step (14bit, 4x oversampling)

Pipelined ADC: 1b/stage or m-bit/stage?

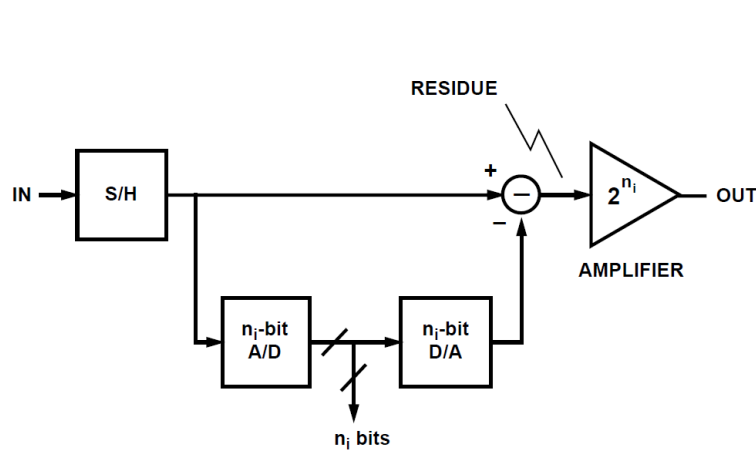
- There was the claim that 1b/stage is better than m-bit/stage



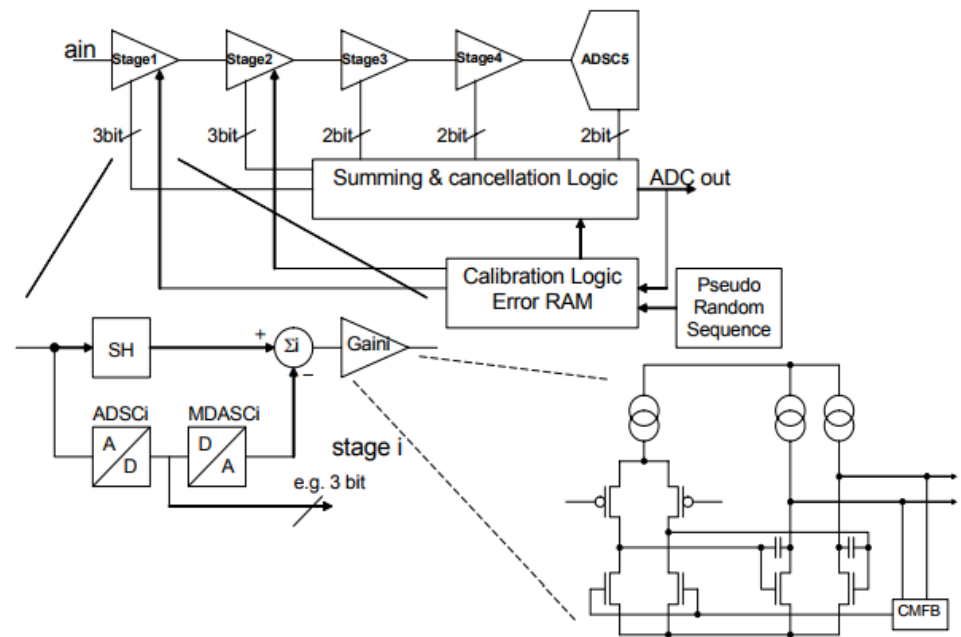
Karanicolas, 15bit, 1Msps pipelined ADC in 2.4um BiCMOS, JSSC'93

Pipelined ADC: 1b/stage or m-bit/stage?

- There was the claim that 1b/stage is better than m-bit/stage
- However, most state-of-the-arts pipelined ADCs use the multi-bit/stage architecture



B. A. Wooley

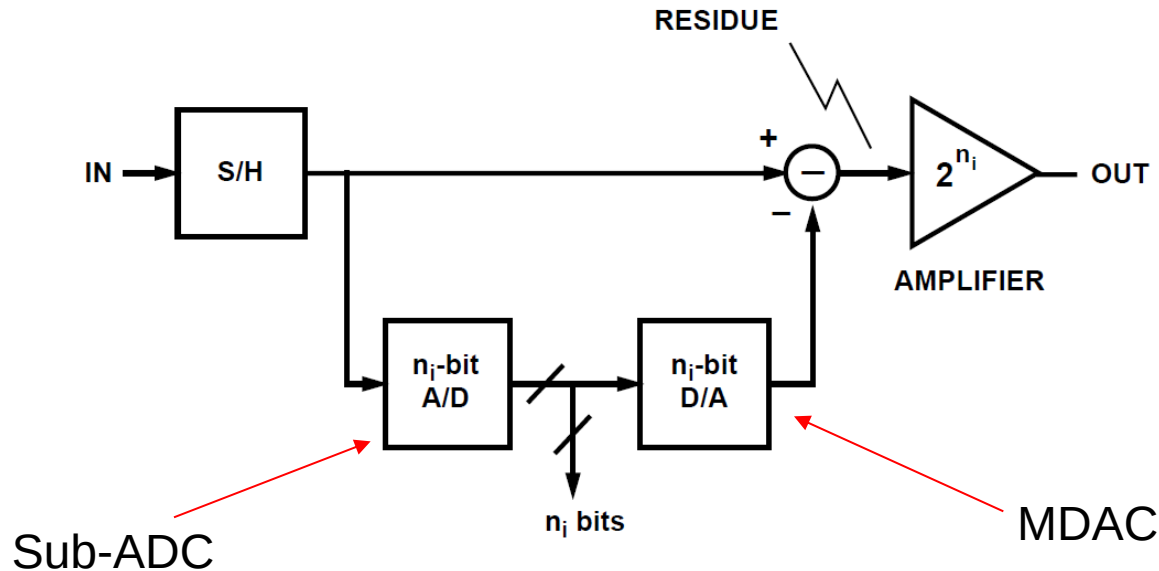


Bogner (Infineon), 14bit, 100Msps pipelined ADC in 0.13um CMOS, ISSCC'06

Similar parts: ADI AD9446 16 bit, 100Msps (0.35um BiCMOS)

Linear LTC2208 16bit, 130Msps (0.35um CMOS)

Hybrid Pipelined



B. A. Wooley, Course Handout

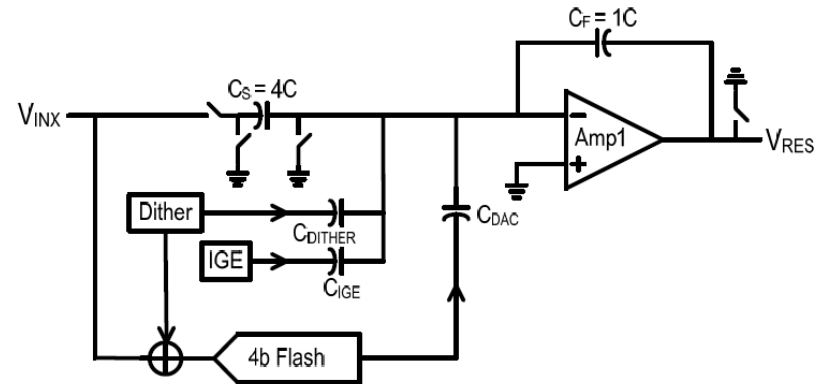
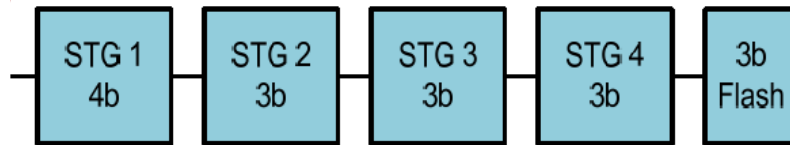
- ❑ Sub-ADC is flash → Pipelined-Flash hybrid (actually just call it pipelined)
- ❑ Sub-ADC is SAR ADC → Pipelined-SAR hybrid



- Pipelined ADC core: 13bit 500Msps
- 3 cascaded SAR stages with 2 bits redundancy
- 5 bits per SAR stage to maximize speed

Pipelined ADC Cores in State-of-the-Art

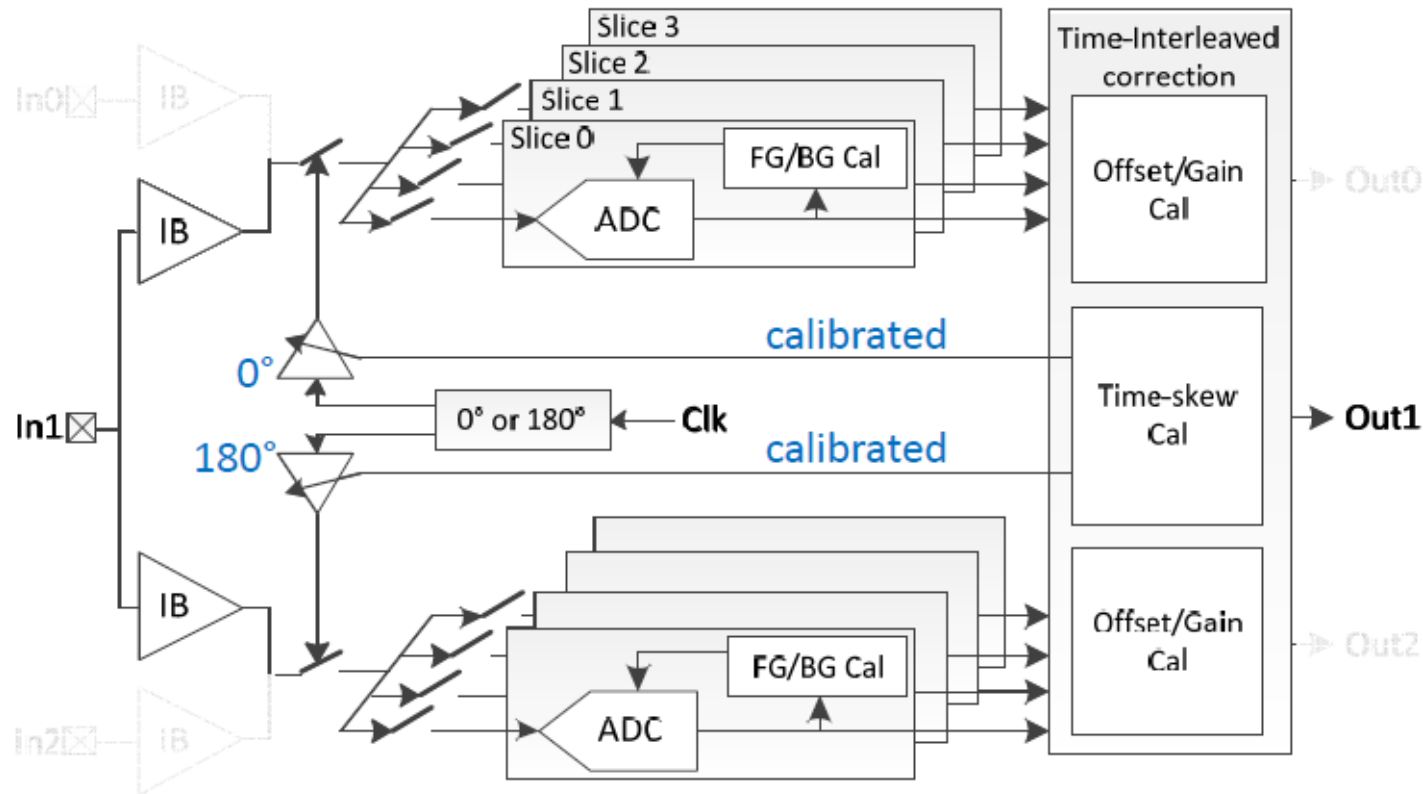
High Speed/High Resolution ADCs (2)



Devarajan, (ADI) A 12b 10GS/s Pipelined ADC in 28nm, ISSCC'17

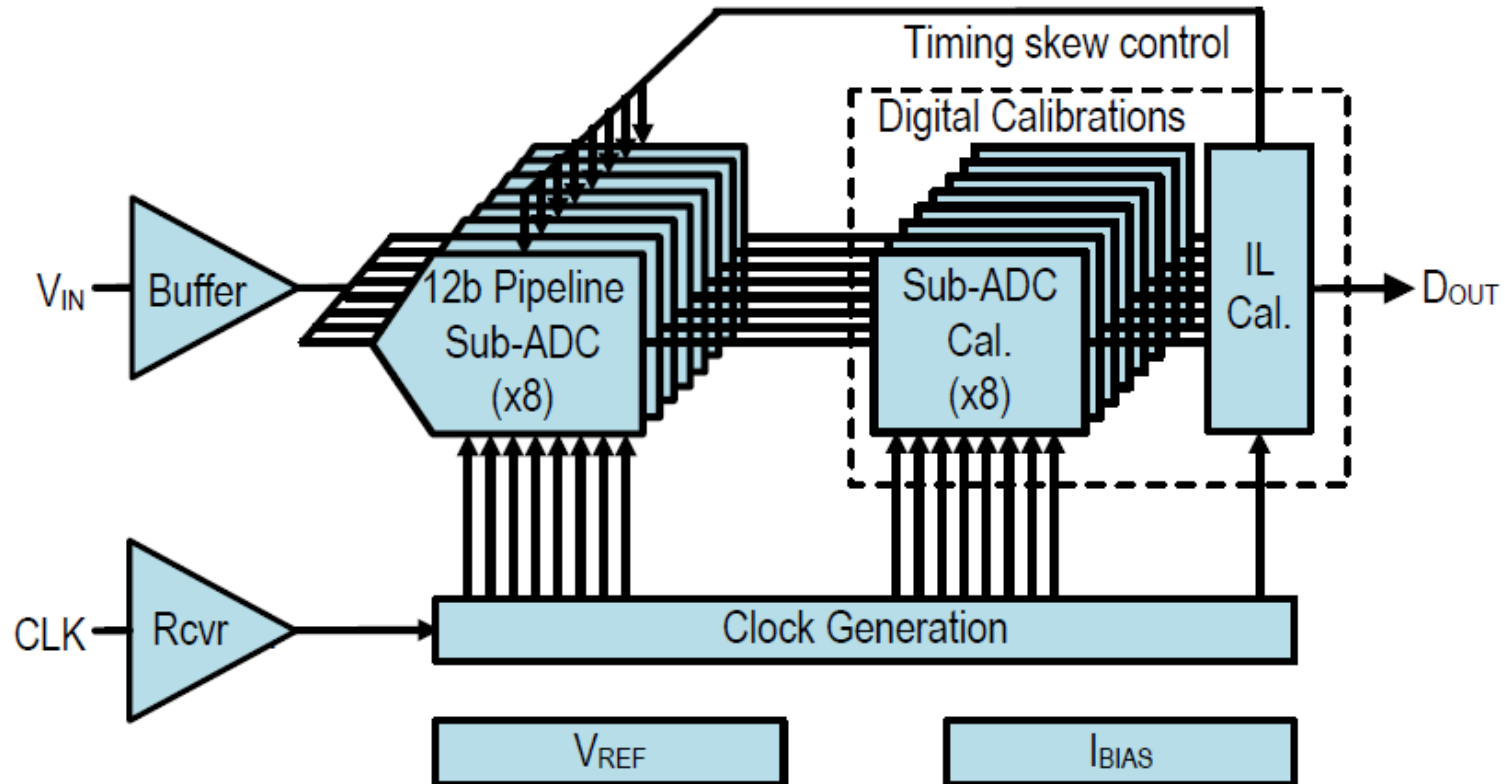
- Pipelined ADC core: 12bit 1.25Gsps
- Each stage has 1-bit redundancy

Interleaving for High Speed Pipelined ADCs (1)



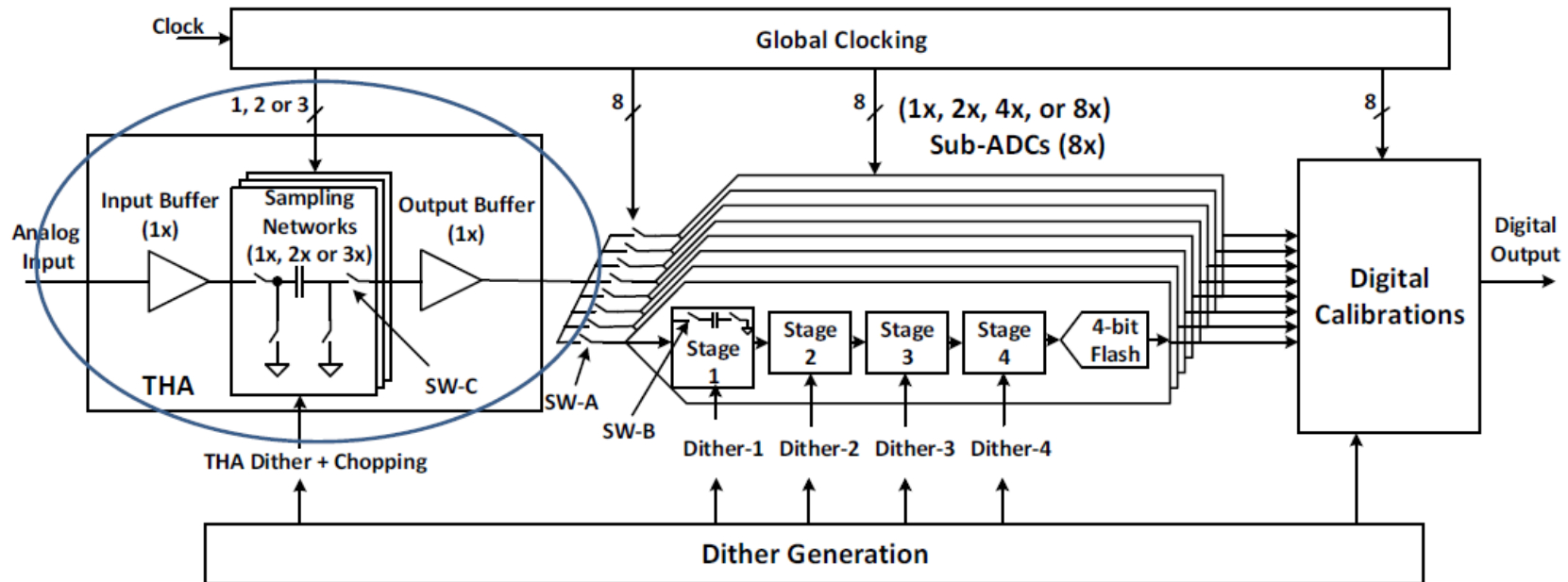
Vaz, (Xilinx) A 13b 4GS/s Pipelined-SAR ADC in 16nm, ISSCC'17

Interleaving for High Speed Pipelined ADCs (2)



Devarajan, (ADI) A 12b 10GS/s Pipelined ADC in 28nm, ISSCC'17

Interleaving for High Speed Pipelined ADCs (3)



Implemented in 16nm FinFET

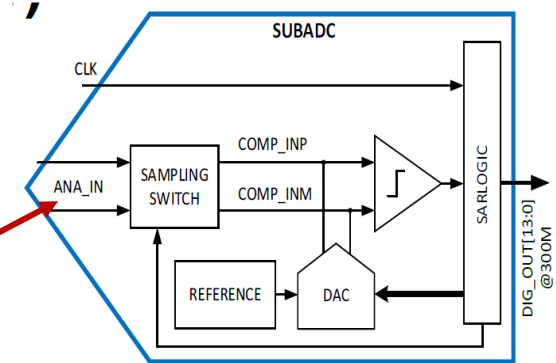
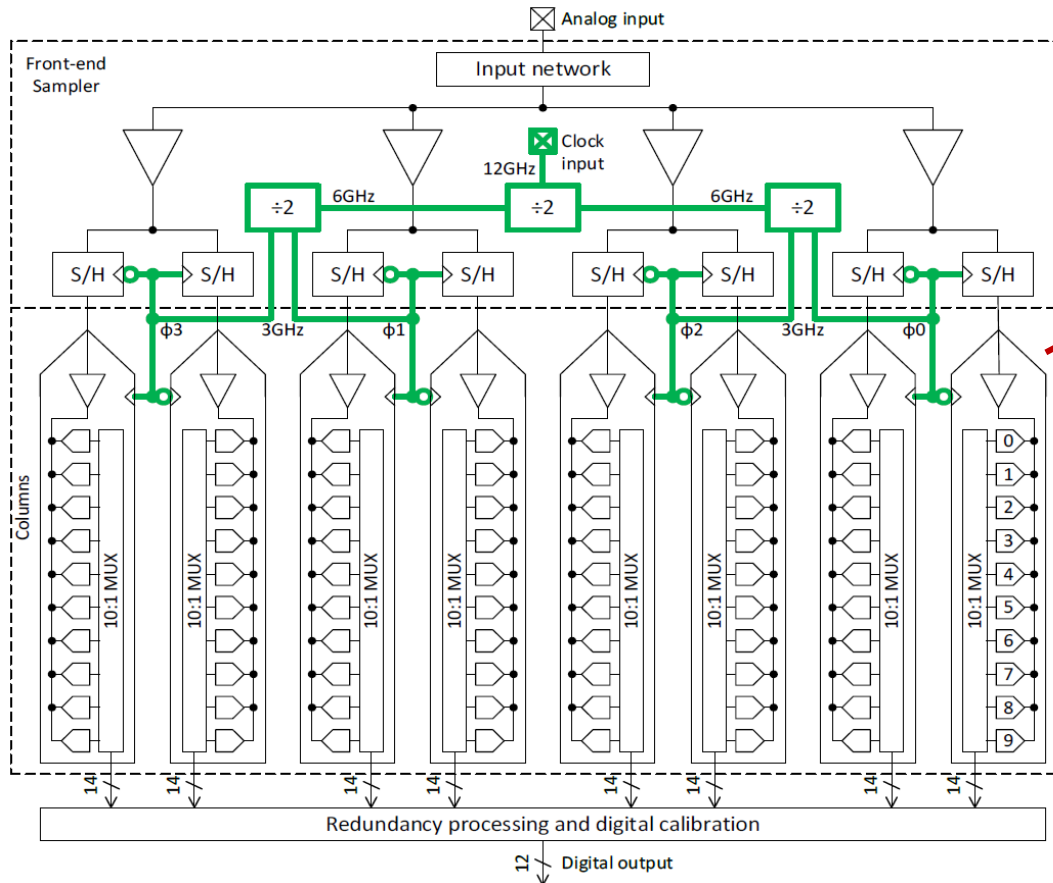
Ahmed, (ADI) 12b 18GS/s RF Sampling ADC, ISSCC'20



High Speed Pipelined ADC Design Challenges

- Clock jitter
- Cross-talk interference
- Mismatch between sub-ADCs
- Output data rate is very high
- Highly dependent on the process → cost!

Time-Interleaved SAR ADC May Also Replace Time-Interleaved Pipelined ADC for High Resolution

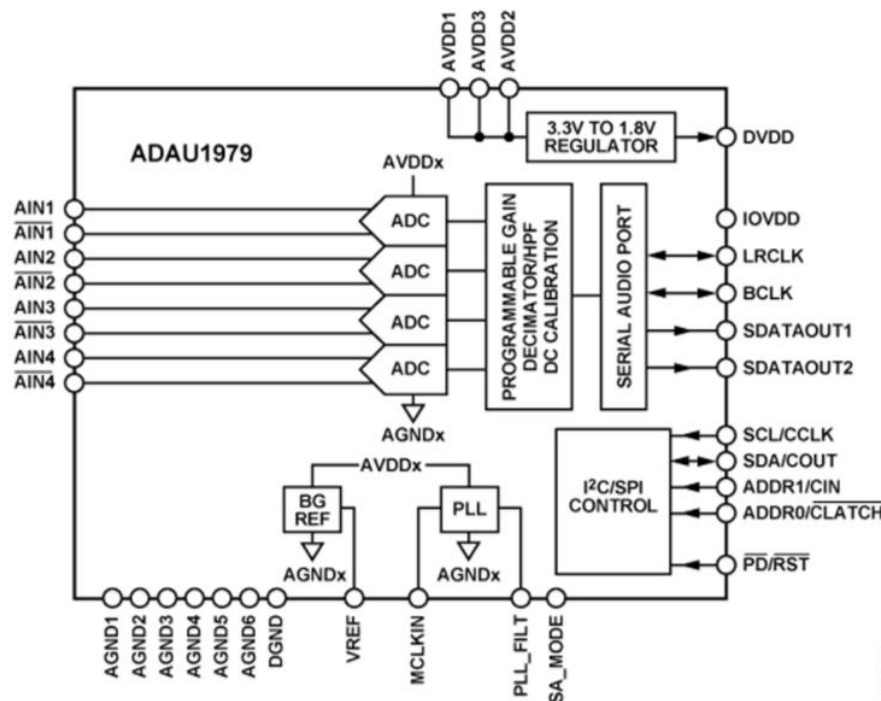


- 7nm FinFET process
- Overall area : 0.9mm²
- Smallest, while fastest 12b ADC!
- Overall power : 750mW

Kumar, A 750mW 24GS/s 12b Time-Interleaved ADC for Direct RF Sampling in Modern Wireless Systems, ISSCC'23

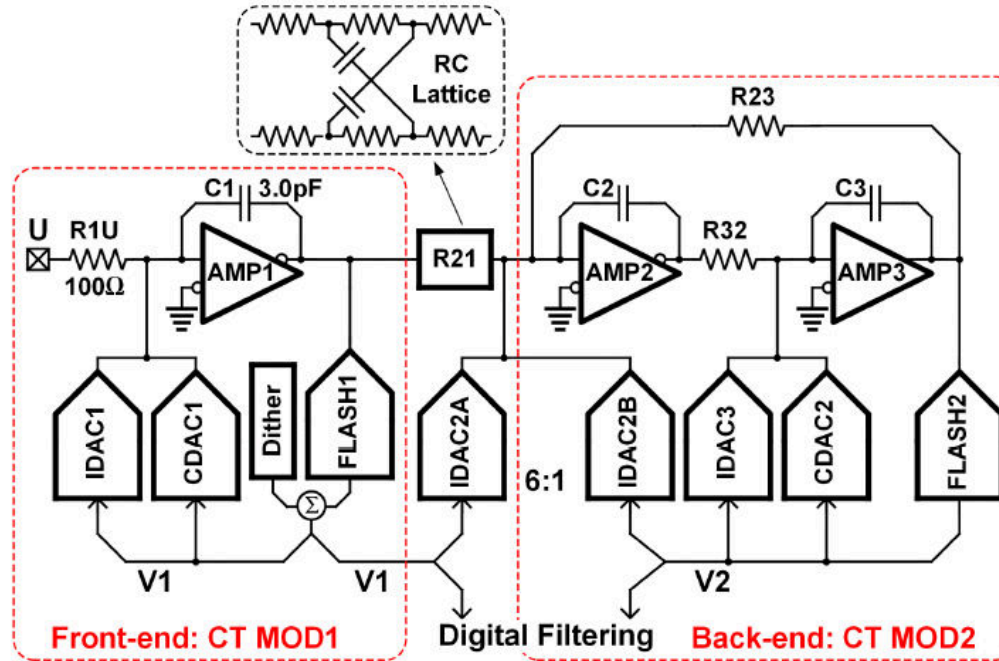
Classical $\Sigma\Delta$ ADC

- $\Sigma\Delta$ ADC is classically for low frequency applications, such as audio applications
- Typical product: ADI ADAU1979, 24bit resolution, 8-192kHz sampling rate, THD+N -95 dB, dynamic range 109dB,



ADI ADAU1979

$\Sigma\Delta$ ADC for High Speed



Dong (ADI), et. al., A 930mW CT 1-2 MASH ADC Achieving 69dB-DR over 465MHz-BW in 28nm CMOS, ISSCC'16

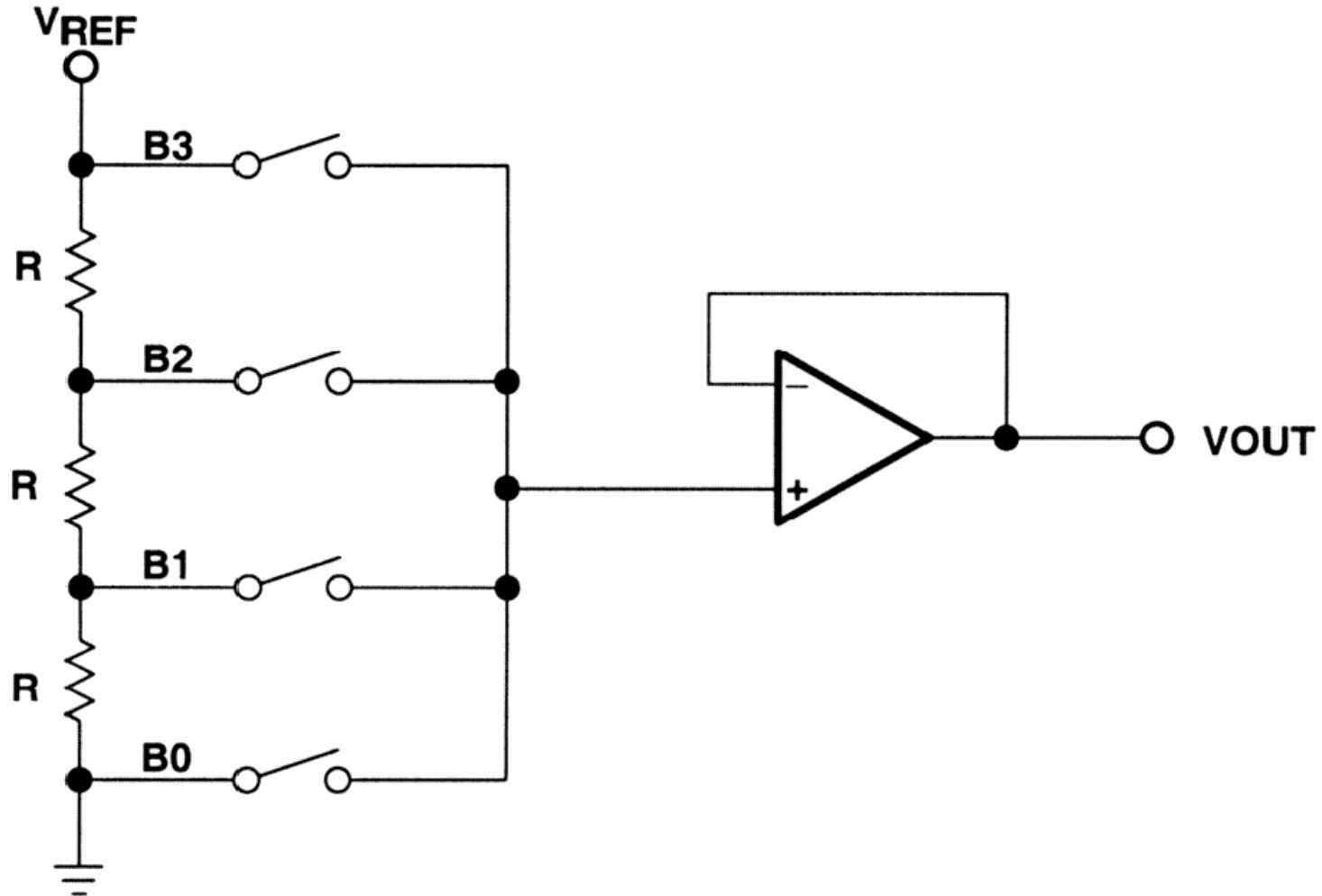
- ❑ continuous-time (CT) multi-stage noise-shaping (MASH)
 - ❑ CT vs. DT
- ❑ Sampling rate 8 GHz, signal bandwidth of 465MHz
- ❑ SNDR 67dB
- ❑ Performance close to the pipeline



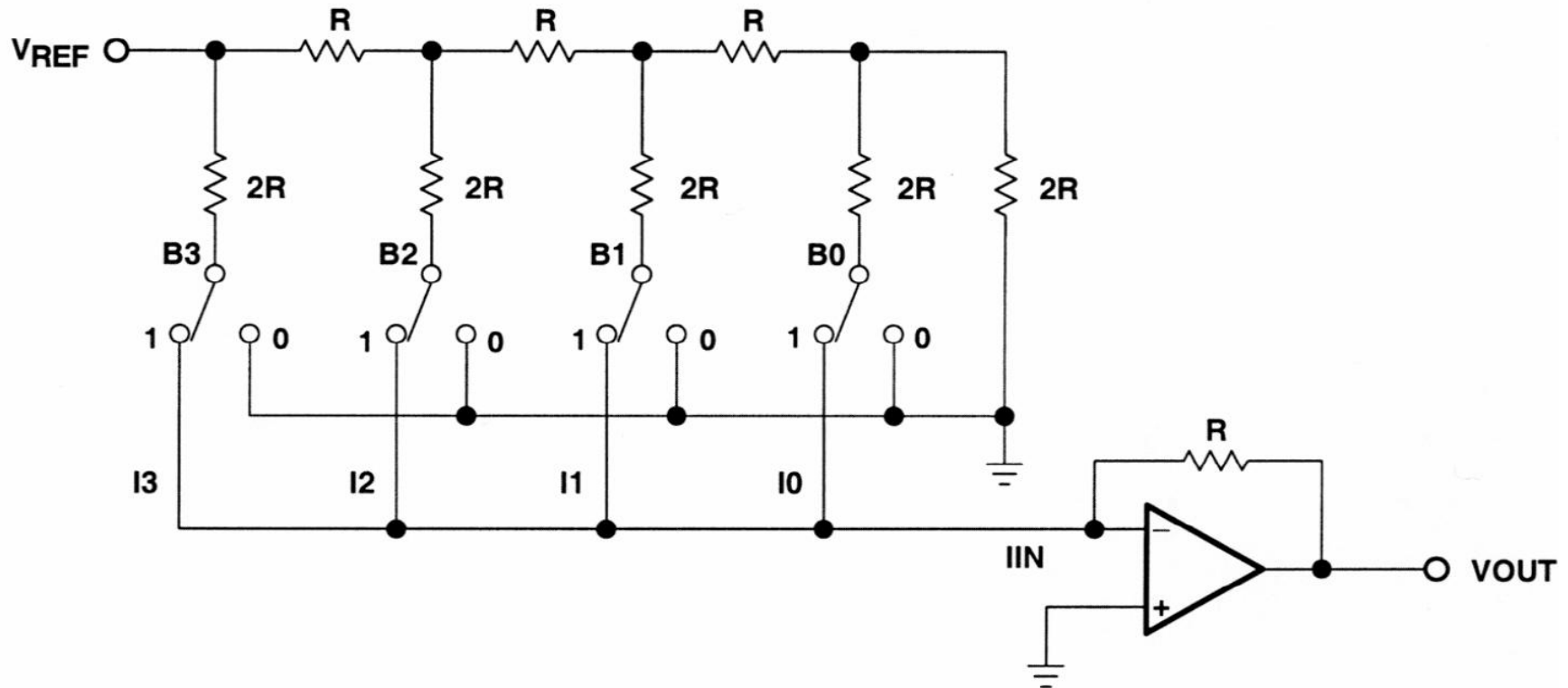
Outline

- Application of ADCs/DACs
- ADC/DAC Performance Spec.
- ADC Architectures
- State-of-the-Art ADCs
- **State-of-the-Art DACs**
- Industry High-Performance ADCs/DACs

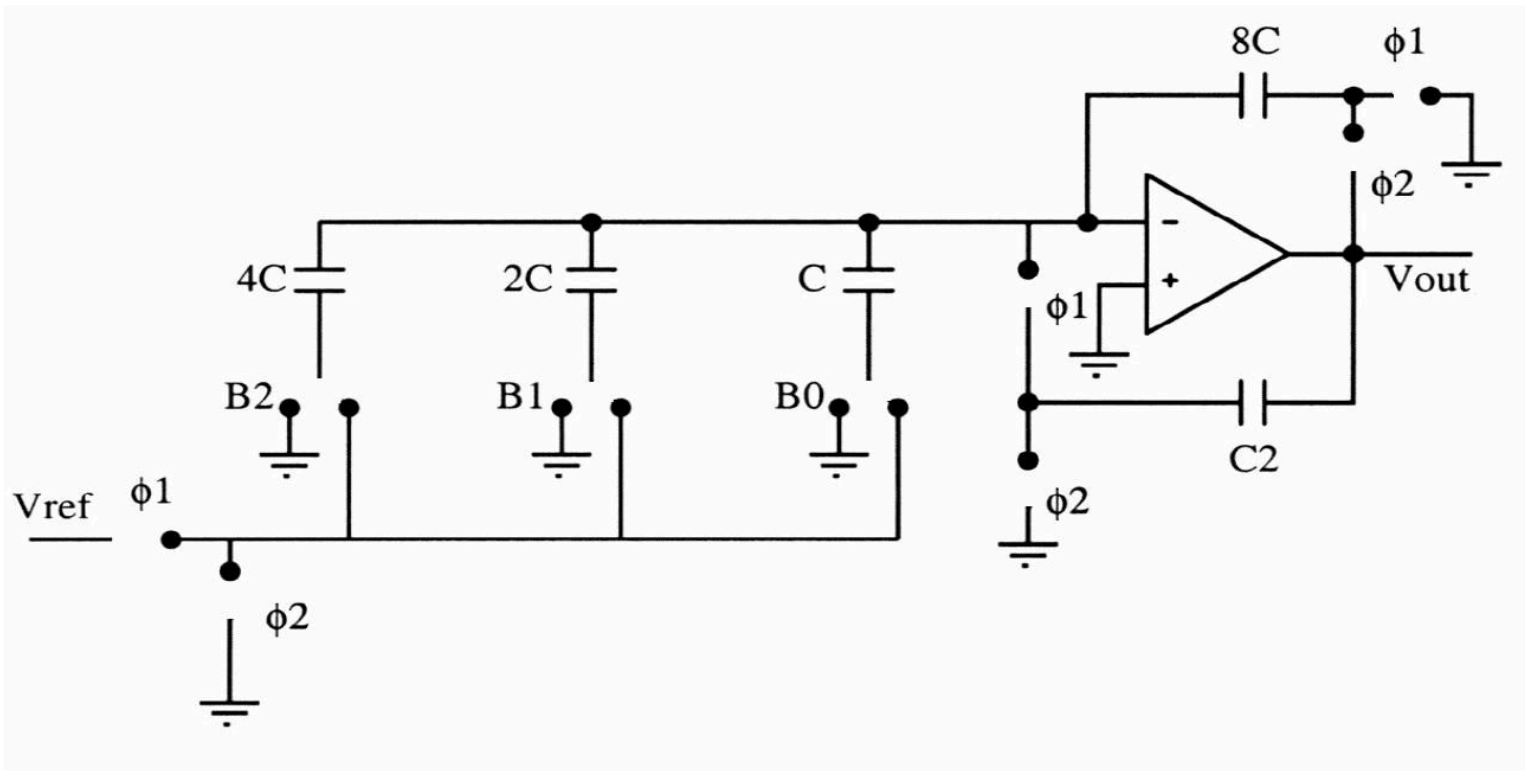
DAC Architecture – R-String



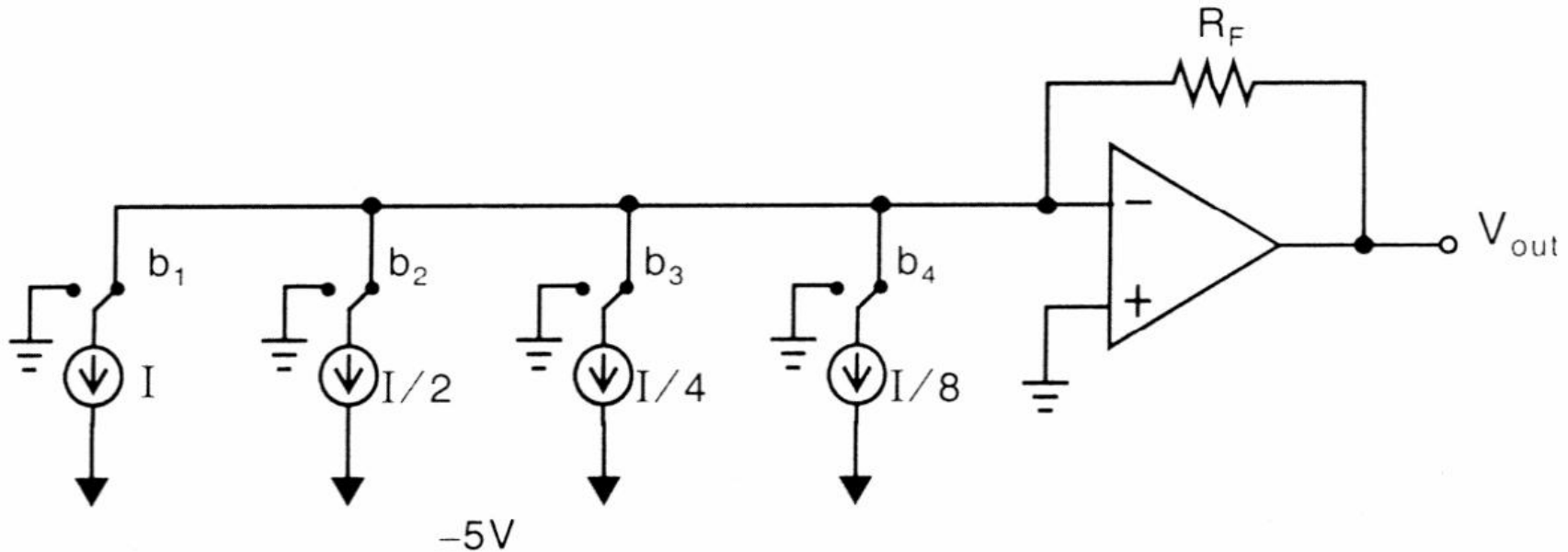
DAC Architecture – R-2R



DAC Architecture – Charge Redistribution

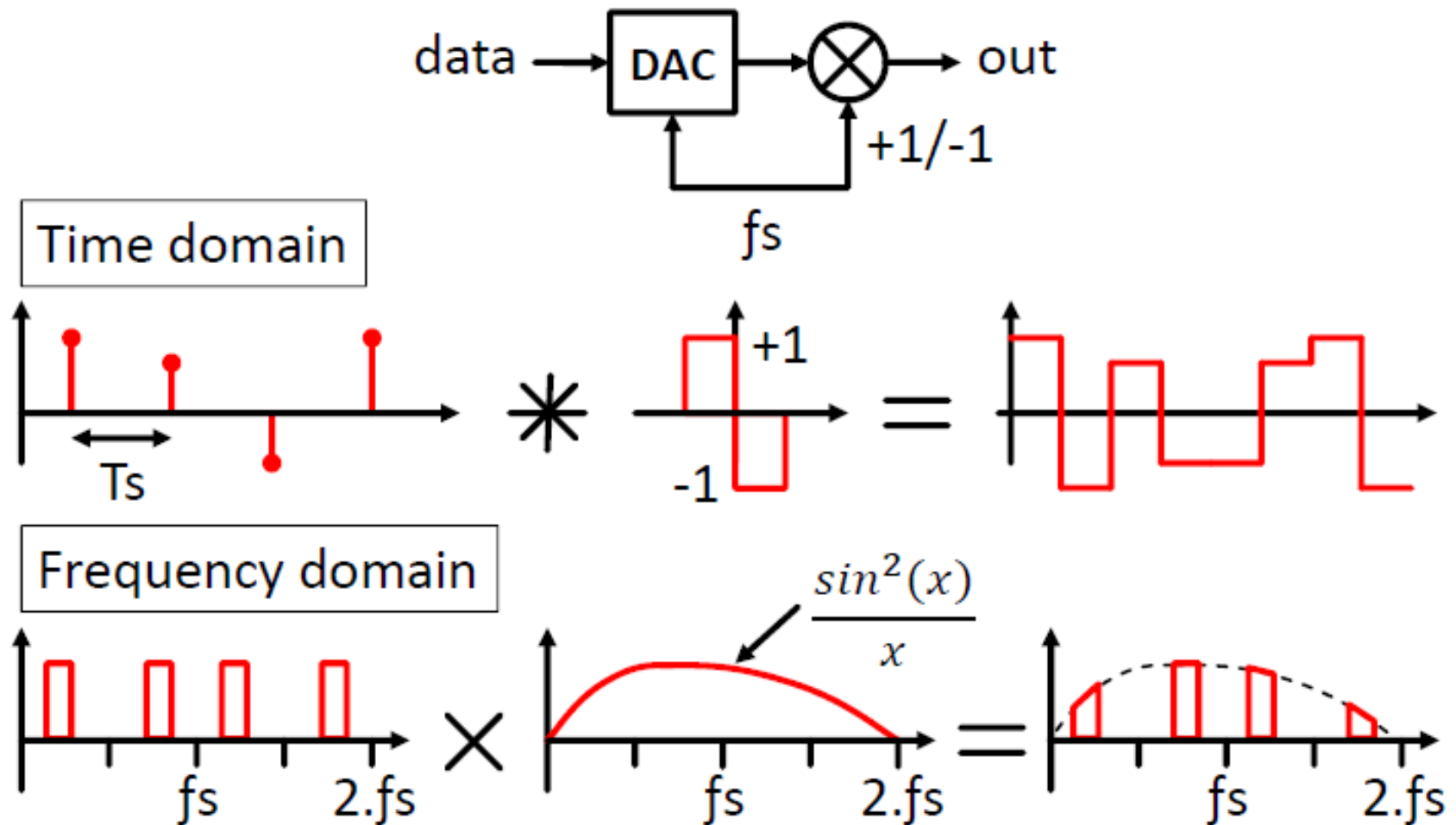


DAC Architectures – Current Steering

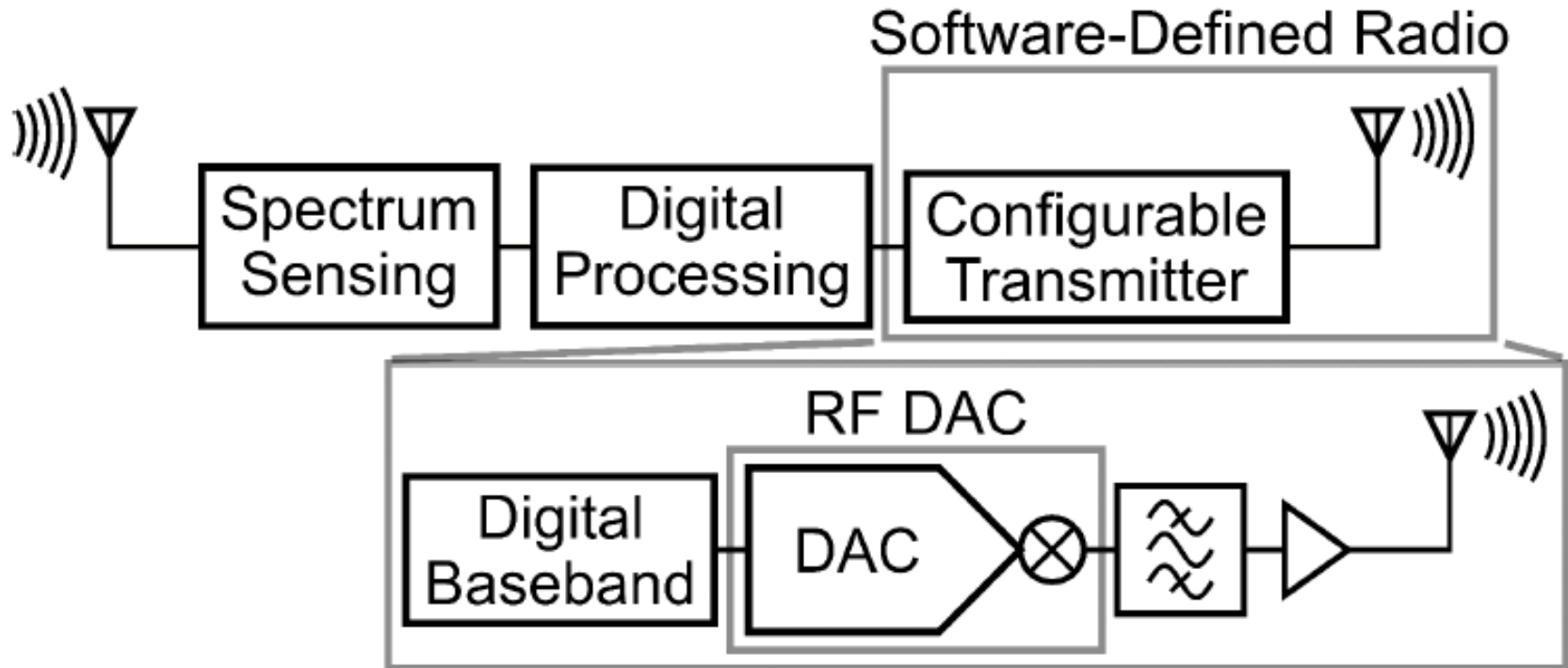


- Fastest with fast current switching

RF DAC (Mixing DAC)



10bit 20Gbps RF DAC

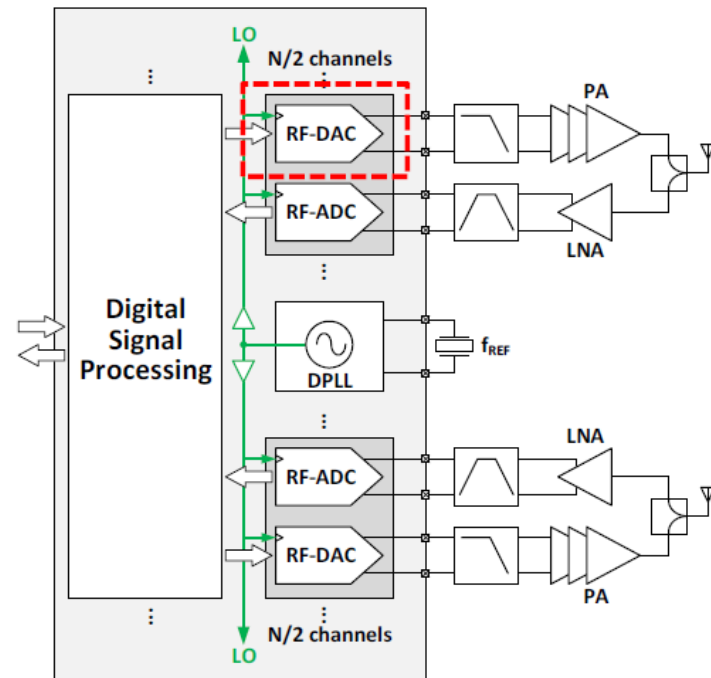


Duncan, et. al. (Xilinx), A 10b DC-to-20GHz Multiple-Return-to-Zero DAC with >48dB SFDR, ISSCC'17

12b 16GS/s RF DAC

Soft-Radio: System Overview

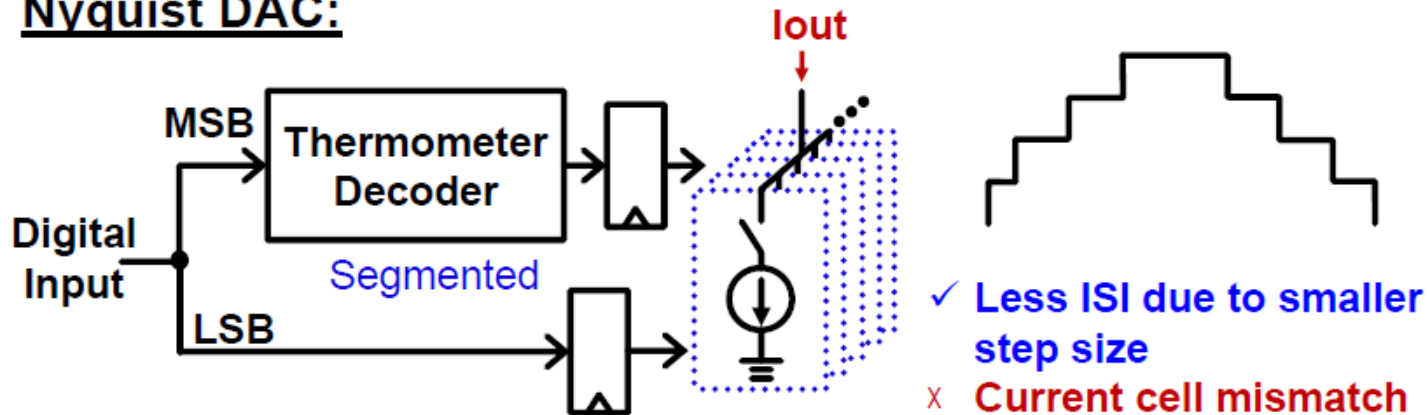
- N-channel TX+RX
- RF-Sampling ADC/DAC with sampling rates of up to 16GSps
- On-chip clock generation (DPLL) [1]
- Integrated DSP functions
- Proposal: Direct RF-Sampling Capacitive DAC for wideband direct RF synthesis
- [1] ISSCC 2021, Paper 32.6



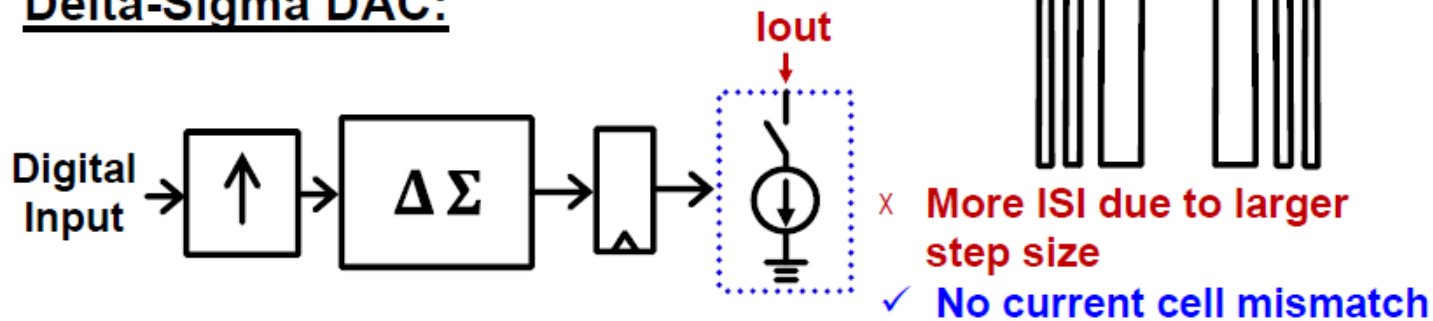
Daniel Gruber, et. al. (Intel), A 12b 16GS/s RF-Sampling Capacitive DAC for Multi-Band Soft-Radio Base-Station Applications, ISSCC'21

DAC: Nyquist vs. $\Delta\Sigma$

Nyquist DAC:

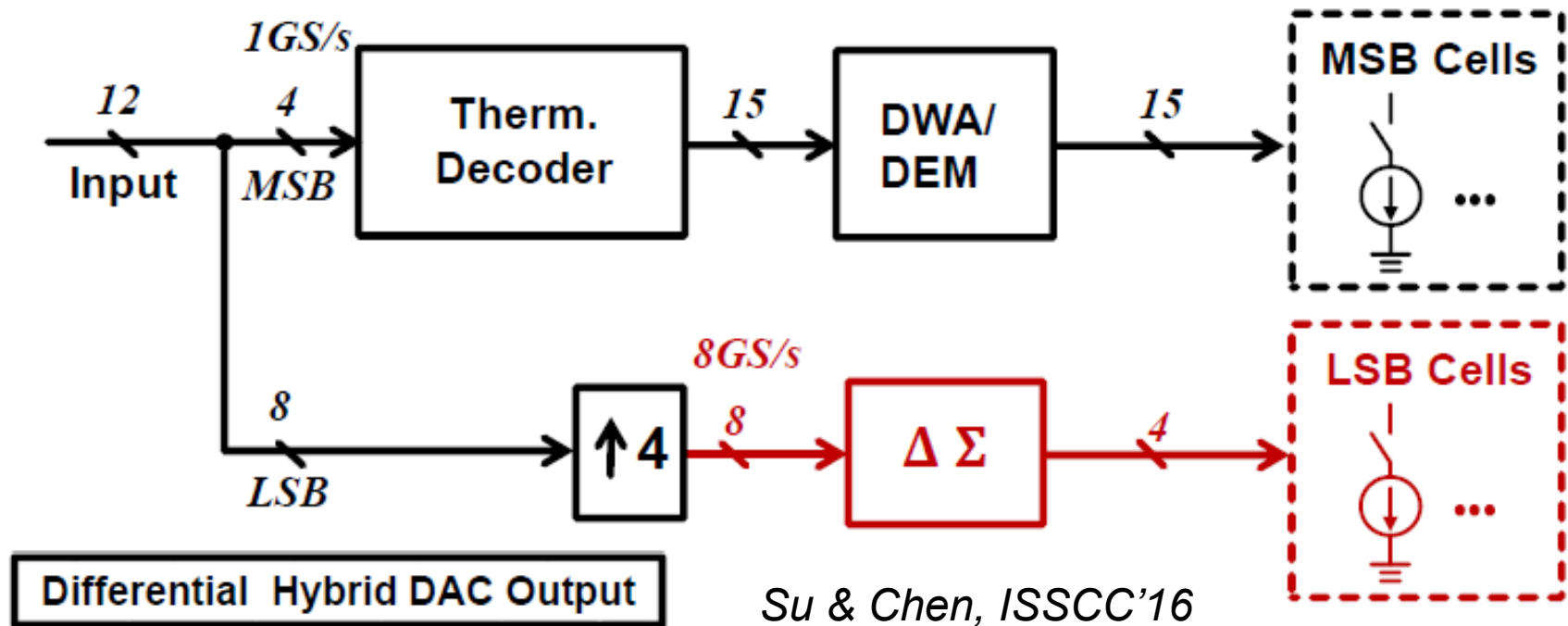


Delta-Sigma DAC:



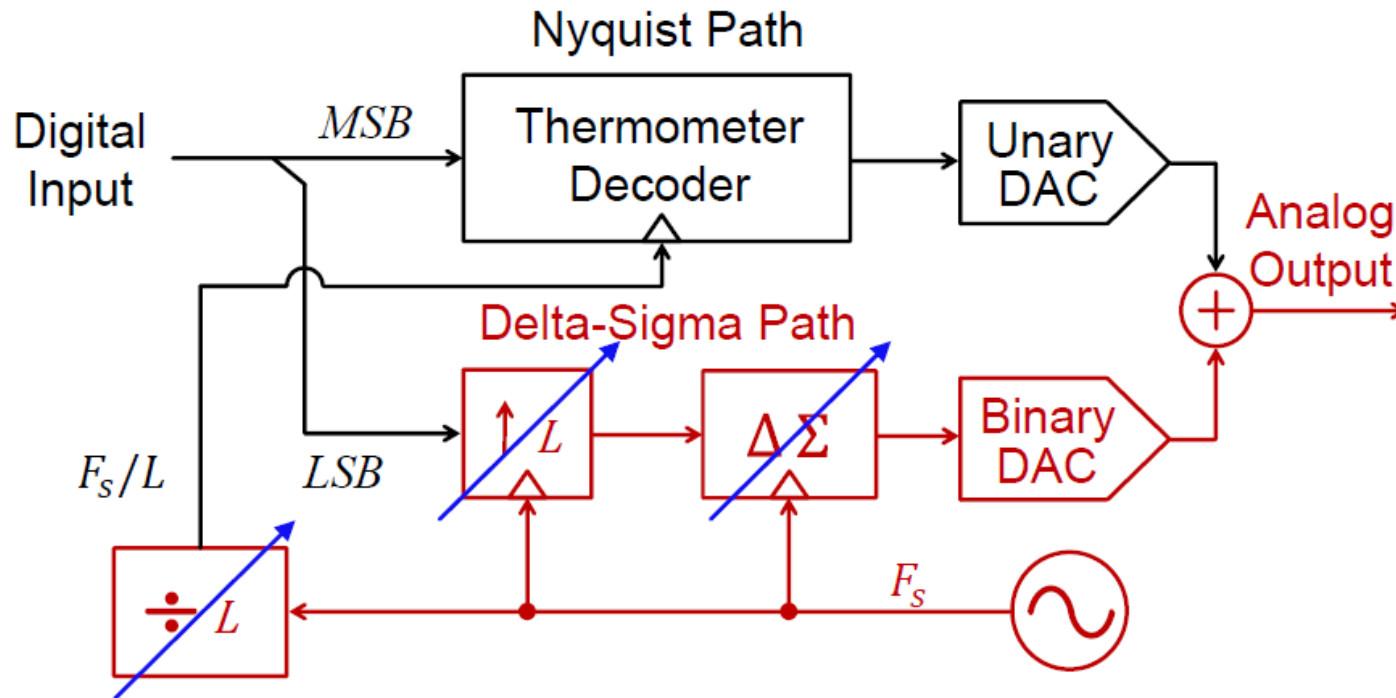
Su & Chen, A 12b 2GS/s Dual-rate Hybrid DAC with Pulsed Timing Error Pre-distortion and In-band Noise Cancellation Achieving >74dBc SFDR up to 1GHz in 65nm CMOS, ISSCC'16

Recent Published DAC: Dual Rate Hybrid



- ❑ Nyquist DAC for MSB: high speed
- ❑ $\Delta\Sigma$ DAC for LSB: high linearity
- ❑ 12bit, 2Gsps, 65nm CMOS, 16mA current

Highest DR*Sampling Rate DAC



- ❑ High-speed tunable bandpass DSM -- Successive Pipelined DSM
- ❑ High-linearity unary DAC -- Inverse-Sinc shaped timing error pre-distortion

Su & Chen, A 16b 12GS/s Single/Dual-Rate DAC with Successive Bandpass Delta-Sigma Modulator Achieving $<-67\text{dBc}$ IM3 within DC-to-6GHz Tunable Passbands, ISSCC'18



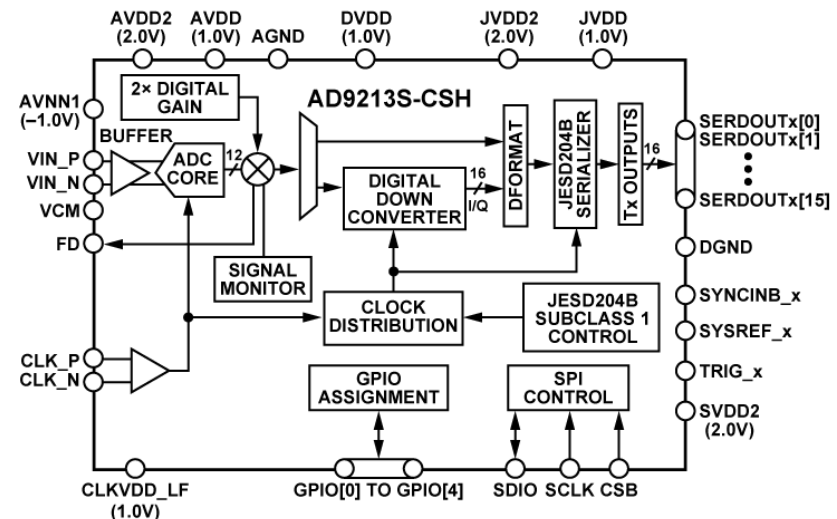
Outline

- Application of ADCs/DACs
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- ADC Architectures
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- State-of-the-Art DACs in Literature
- **Industry High-Performance ADCs/DACs**

ADI AD9213S

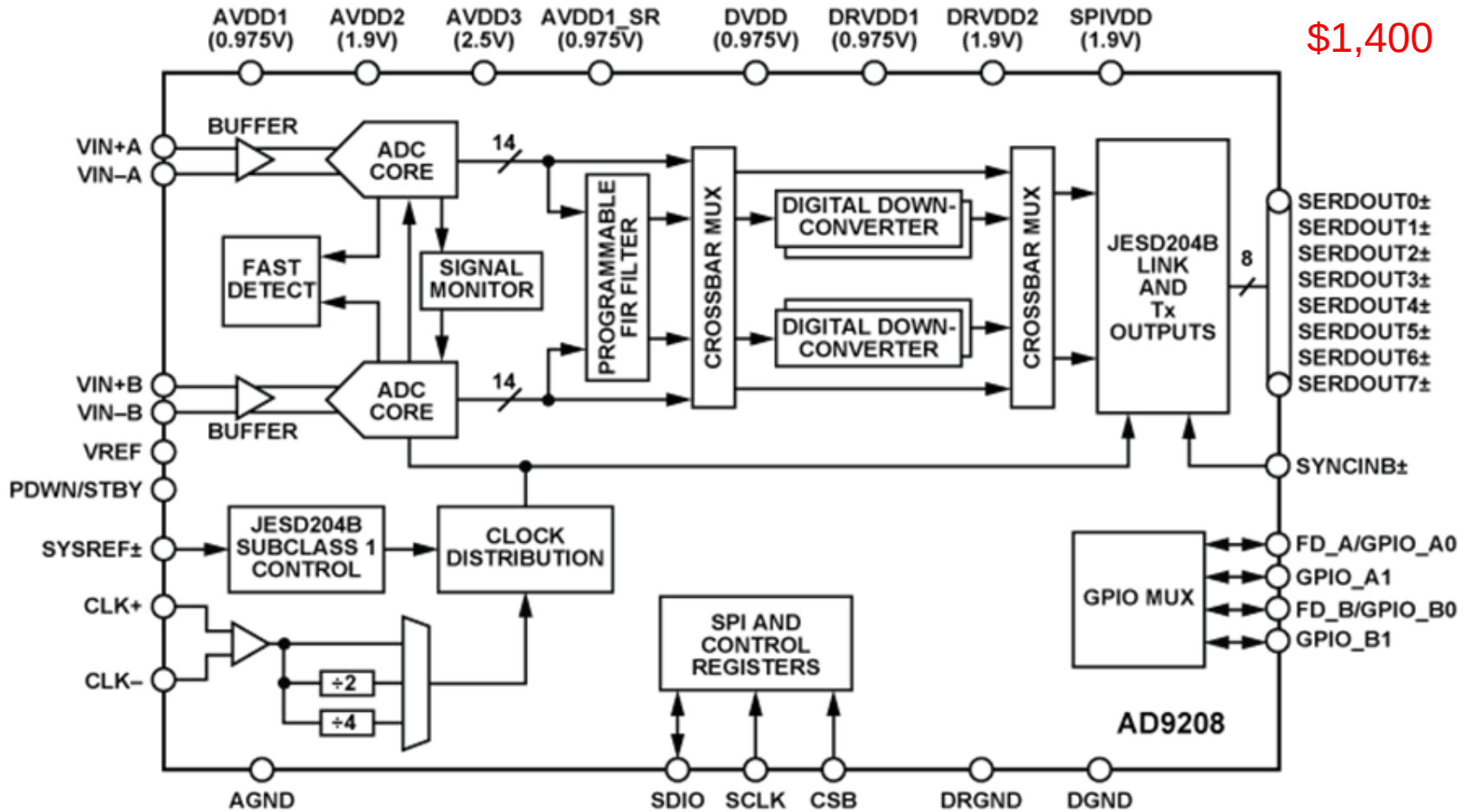
RF ADC 12-Bit, 6 GSPS/10.25 GSPS

- Noise Spectral Density (NSD)
 - 155 dBFS/Hz at 10 GSPS with -9 dBFS, 170 MHz input
 - 153 dBFS/Hz at 10 GSPS with -1 dBFS, 170 MHz input
- SFDR: 70 dBFS at 10 GSPS with 1000 MHz input
- Low power dissipation: <4.6 W typical at 10 GSPS
- Unit price
 - ADC9213S, \$21089 (aerospace)
 - ASC9213, \$1171



ADI AD9208 14-Bit, 3 GSPS

\$1,400

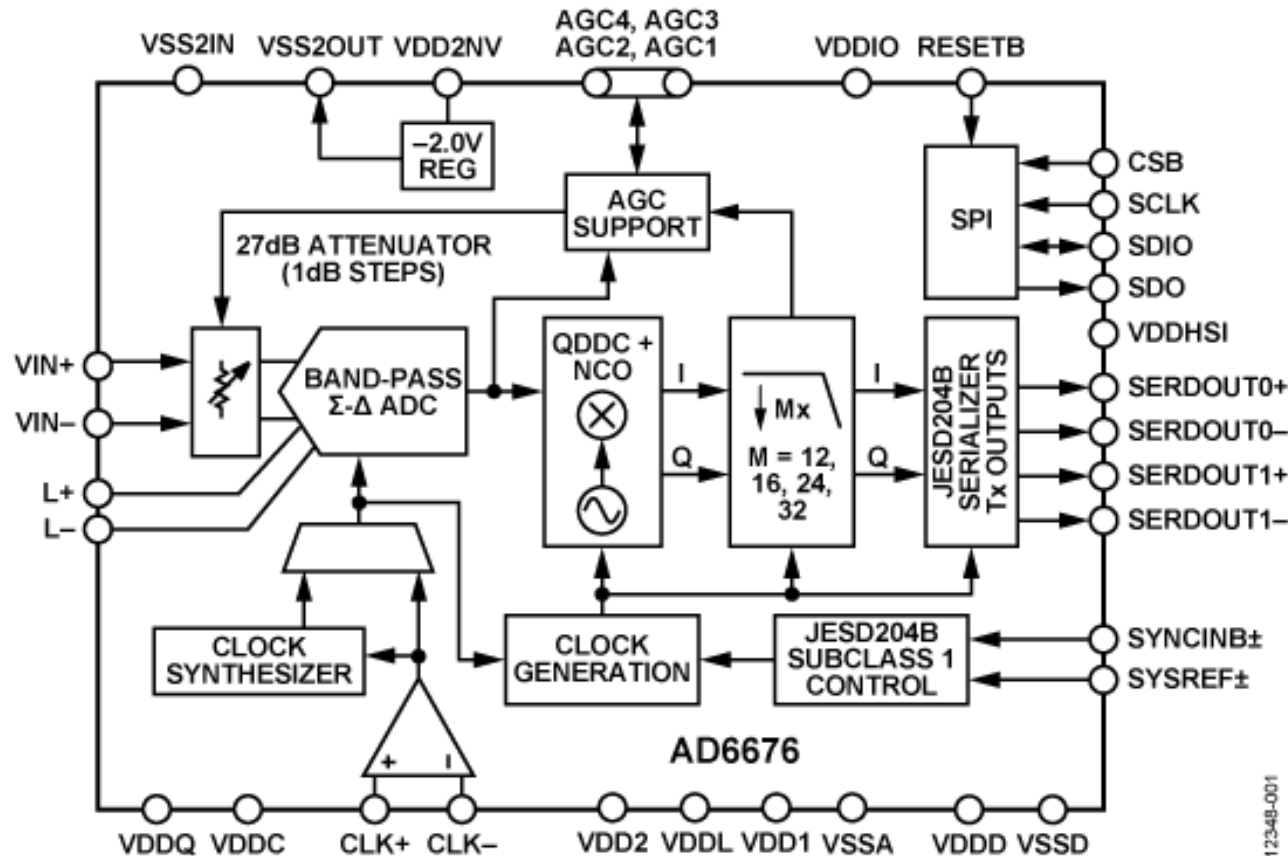


15547-001

ADI AD9208 14-Bit, 3 GSPS

ADI AD6676 16-Bit, 3.2 GSPS

\$168

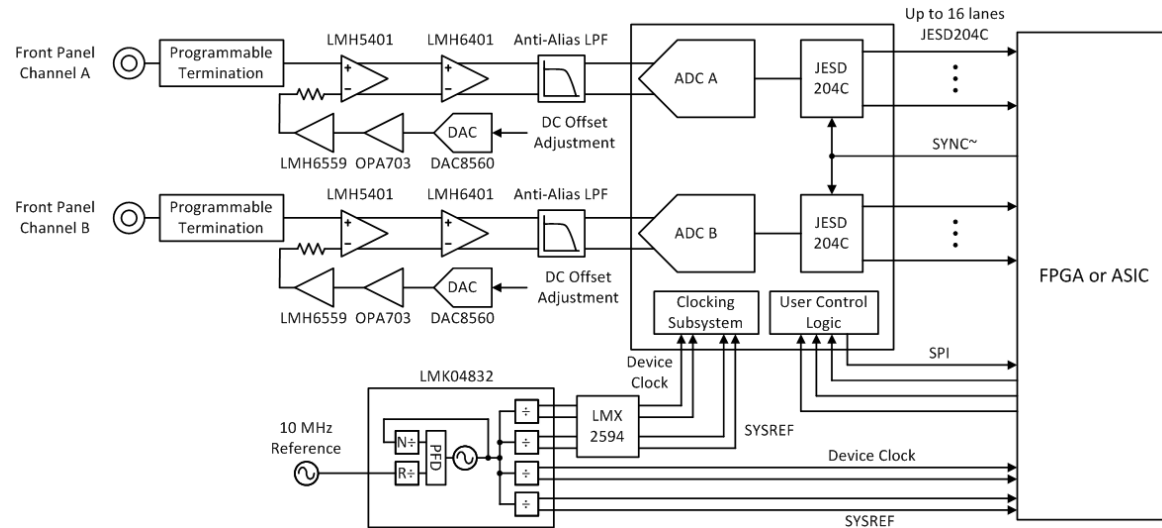
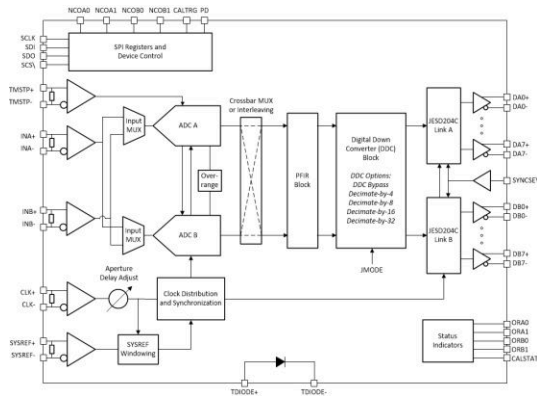


- Tunable band-pass Σ - Δ ADC
- 20 - 160 MHz signal bandwidth

ADI AD6676 16-Bit, 3.2 GSPS

TI ADC12DJ5200

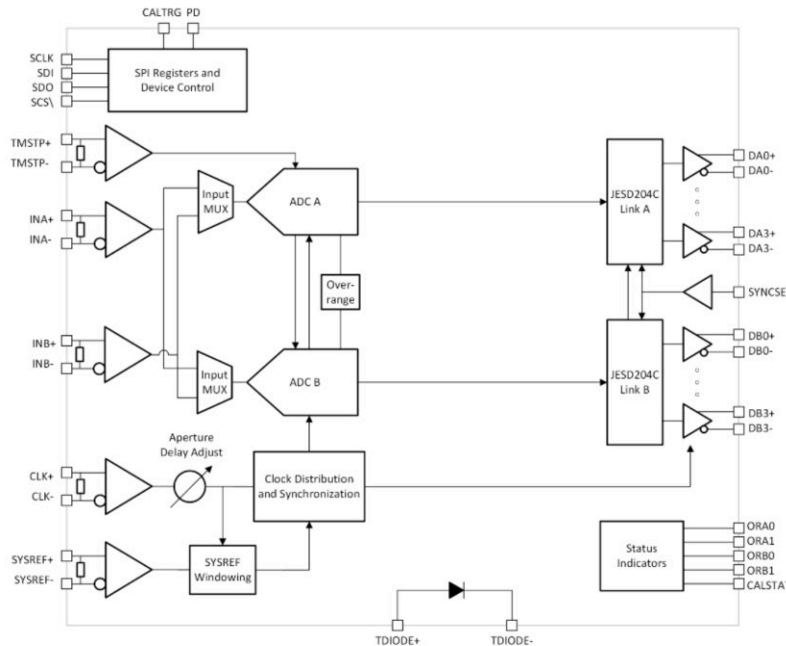
RF-sampling 12-bit ADC 10.4GSPS



- 12-bit resolution
- Up to 10.4GSPS in single-channel mode (sampling at both rising/falling edge)
- Up to 5.2GSPS in dual-channel mode
- Analog input bandwidth (−3dB): 8GHz
- Usable input frequency range: > 10 GHz
- Full-scale input voltage (VFS, default): 0.8VPP
- Power consumption: 4W, Power supplies: 1.1V, 1.9V
- Unit price: \$2,777 (1ku)

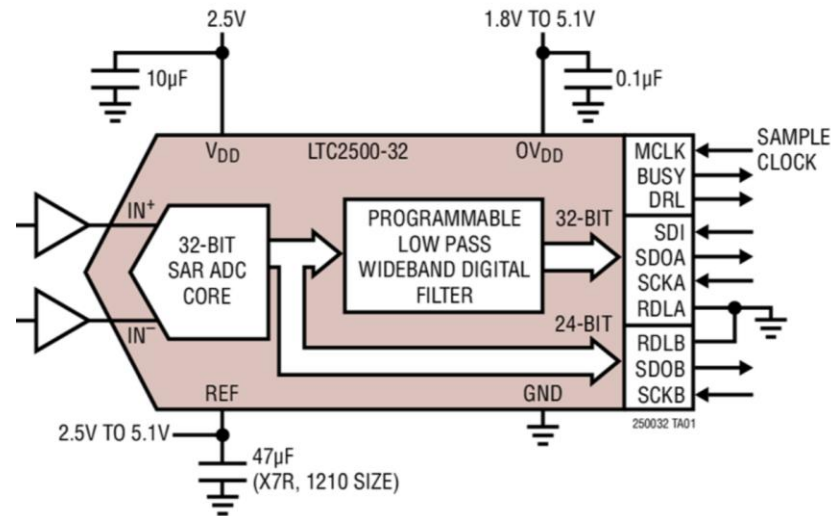
RF-sampling 8-bit ADC 10.4GSPS

- 8-bit resolution
- Up to 10.4 GSPS in single-channel mode
- Up to 5.2 GSPS in dual-channel mode
- Applications
 - Oscilloscopes and wideband digitizers
 - Communications testers
 - Electronic warfare (SIGINT, ELINT)
 - Satellite communications (SATCOM)
 - RF-sampling software-defined radio (SDR)
 - Spectrometry



ADI LTC2500-32 32-Bit Over-Sampling ADC

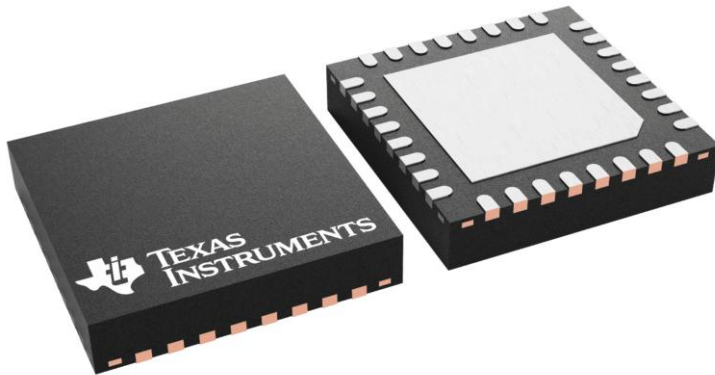
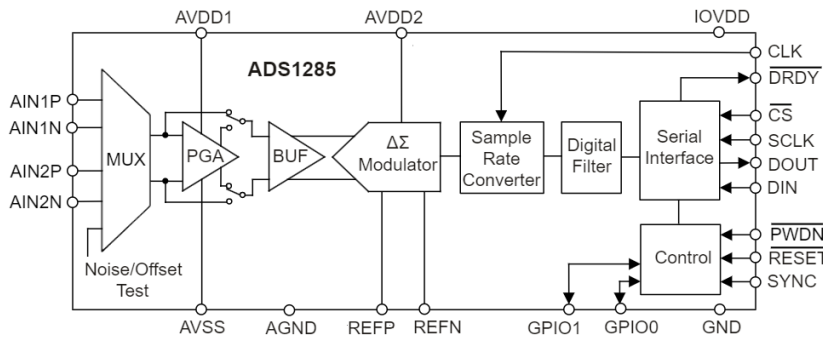
- $\pm 0.5\text{ppm}$ INL (Typ)
- 104dB SNR (Typ) at 1Msps
- 148dB Dynamic Range (Typ) at 61sps
- Guaranteed 32-Bit No Missing Codes
- Dual Output 32-Bit SAR ADC
 - 32-Bit Digitally Filtered Low Noise Output
 - 24-Bit Differential + 7-Bit Common Mode 1Msps Output with Overrange Detection
- Guaranteed Operation to 85°C
- Low Power: 24mW at 1Msps
- Unit price: \$42.38



TI ADS1263



32-bit 38-kSPS delta-sigma ADC

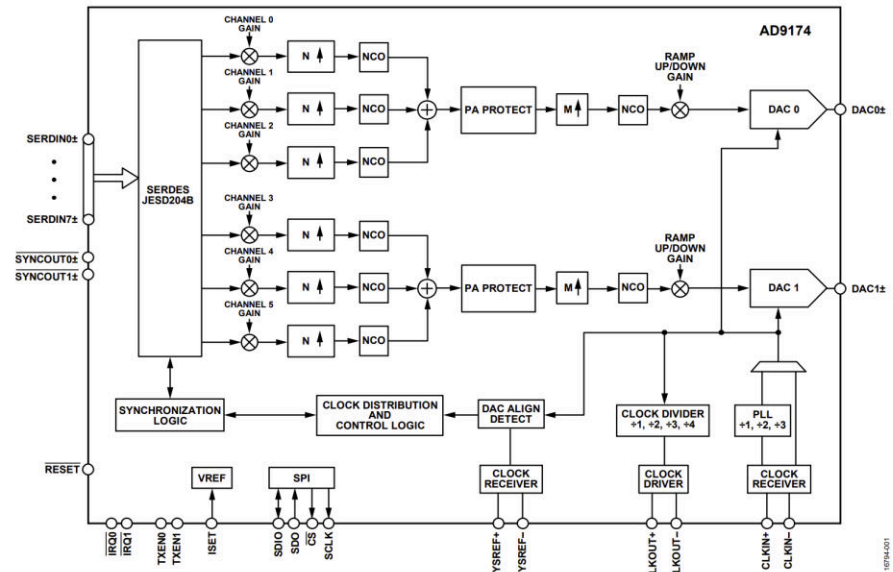


- Precision, 32-bit, $\Delta\Sigma$ ADC
- Data rates: 2.5 SPS to 38400 SPS
- Offset drift: $1 \text{ nV}/^{\circ} \text{C}$
- Gain drift: $0.5 \text{ ppm}/^{\circ} \text{C}$
- Noise: 7 nVRMS (2.5 SPS, gain = 32)
- Linearity: 3 ppm
- – Temperature drift: $2 \text{ ppm}/^{\circ} \text{C}$
- 50-Hz and 60-Hz rejection
- Unit price: \$11 (was \$46)

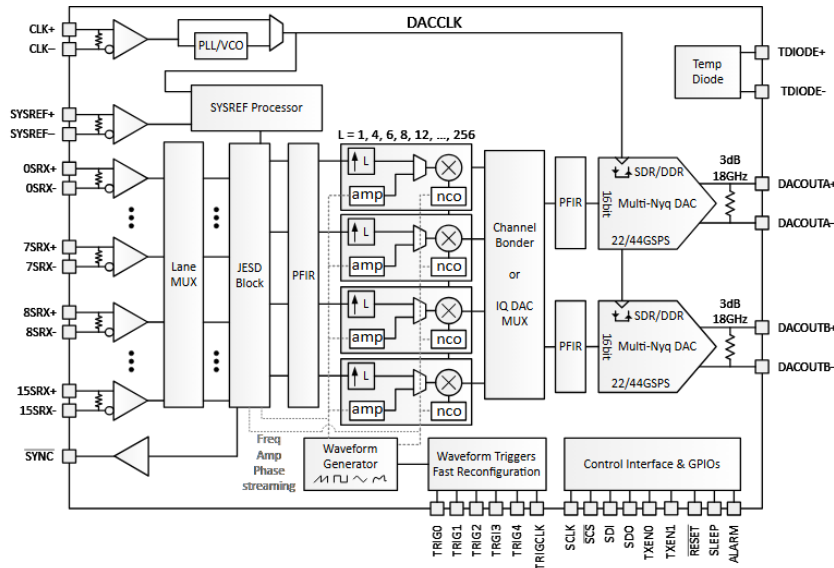
ADI 9174

16-Bit, 12.6 GSPS RF DAC & DDS

- 2-tone IMD3 = -83 dBc at 1.84 GHz, -7 dBFS/tone RF output
 - SFDR < -80 dBc at 1.84 GHz, -7 dBFS RF output
 - Flexible 8-lane, 15.4 Gbps JESD204B interface
 - Final 48-bit NCO that operates at the DAC rate to support frequency synthesis up to 6 GHz
 - 2.54 W with 2 DACs at 12 GSPS
 - Unit Price: \$564.96(1k)
- Wireless communications infrastructure
 - Multiband base station radios
 - Microwave/E-band backhaul systems
 - Instrumentation, automatic test equipment (ATE)
 - Radars and jammers
-
- The diagram illustrates the internal architecture of a high-speed DAC. On the left, the SERDES JESD204B interface is shown with inputs SERDIN0±, SERDIN7±, and SYNCOUT0±, SYNCOUT1±, and a RESET pin. The core consists of six parallel channels (0 to 5). Each channel has a gain stage (CHANNEL 0 GAIN to CHANNEL 5 GAIN) followed by a DAC (N) and a Non-Coherent Oscillator (NCO). The outputs of the DACs are summed in pairs (0+1, 2+3, 4+5) and then pass through PA PROTECT and M blocks. The summed signals are then processed by RAMP UP/DOWN GAIN blocks. The bottom section shows the control logic, including SYNCHRONIZATION LOGIC, CLOCK DISTRIBUTION AND CONTROL LOGIC, DAC ALIGN DETECT, CLOCK RECEIVER, CLOCK DRIVER, and PLL blocks, which manage the timing and frequency of the DACs.



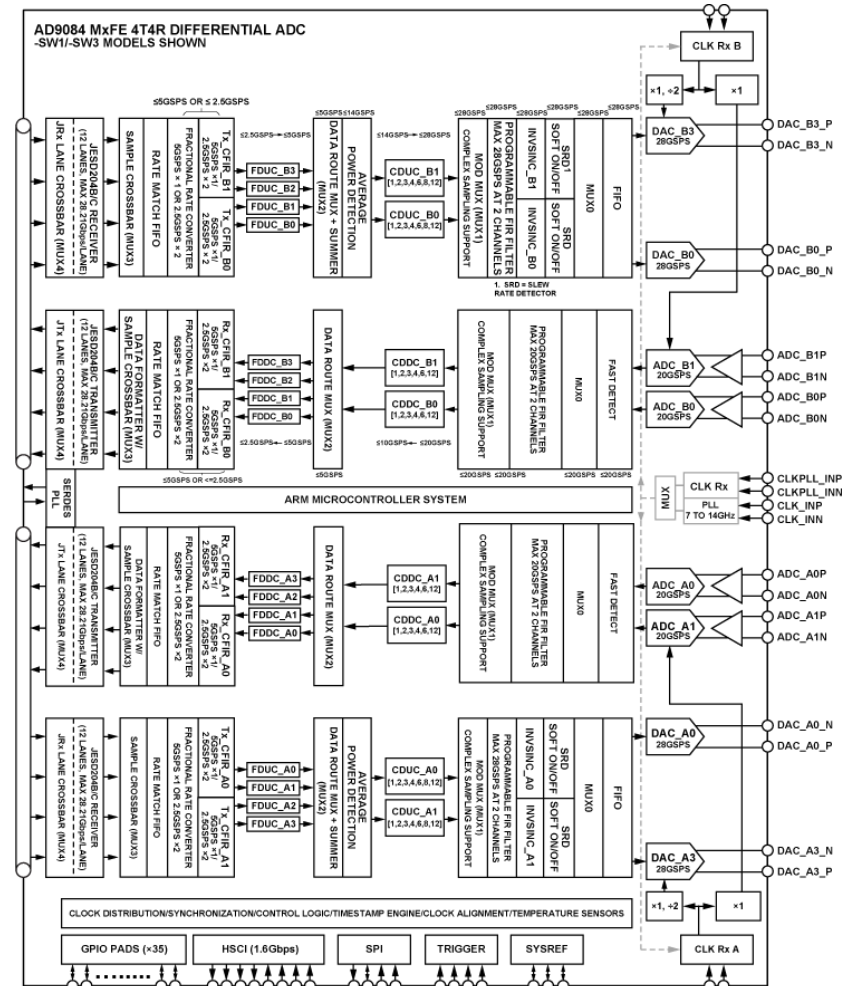
TI DAC39RF20 16-bit 44GSPS RF-sampling DAC



- 16-bit Multi-Nyquist DAC core
- DAC sample rate:
 - NRZ, RF Modes: 22GSPS
 - DES2xL, DES2xH modes: 44GSPS
- Performance at $f_{OUT} = 10\text{GHz}$, DES2xL mode
 - Noise floor (small signal): – 170dBFS/Hz
 - SFDR (-0.1dBFS) : –55dBc
 - IMD3 (-7dBFS) : –60dBc
- Four integrated digital up-converters (DUC)
- Direct digital synthesis (DDS) functions
- Unit Price: \$1230

AD9084 MxFE Quad, 16-Bit, 28GSPS RF DAC and Quad, 12-Bit, 20 GSPS RF ADC

- 4 16-bit RF DACs + 4 12-bit RF ADCs (4T4R)
- Usable RF analog bandwidth to 18GHz
- Maximum DAC/ADC sample rate up to 28GSPS/20GSPS
- Typical power consumption 29.9W
- Unit Price: \$1750



TI AFE8192 Wide-band 16-channel RF transceiver



- 16 RF sampling 12GSPS TX DACs
- 16 RF sampling 6GSPS RX ADCs
- RF frequency range: up to 7.2GHz
- Maximum RF signal bandwidth:
 - TX/FB: 1200MHz
 - 1600MHz in 8-channel mode
 - RX: 600MHz
 - 1200MHz in 8-channel mode
- Unit price: \$2,499



Thanks for Your Attention!

Questions?

Hanjun Jiang

jianghanjun@tsinghua.edu.cn



文献调研作业

- 调研主题: High-speed ADC/DAC techniques in wireline transceivers
- 调研内容: 在最新的高速wireline transceiver设计中, 基于ADC/DAC+DSP的方案应用越来越多, 请阅读近4年ISSCC会议论文中的wireline transceiver, 总结分析: (1) ADC/DAC+DSP的方案wireline transceiver在技术上有什么优劣势? (2) 用在超高速wireline transceiver的ADC/DAC有什么技术特点和性能现状? (3) 用在超高速wireline transceiver的ADC/DAC设计的主要技术挑战和可能的解决方案。
- 报告要求: 阅读文献量不少于8篇, 其中精读不少于4篇, 报告结尾附参考文献, 报告正文字数不少于5000字
- 建议阅读文献:
 - 2025 ISSCC的Session 7: Ultra-High-Speed Wireline 和 Session 36: Ultra-High-Density D2D and High-Performance Optical Transceivers
 - 2024 ISSCC的Session 7: Ultra-High Speed Wireline 和 Session 18: High-Performance Optical Transceivers
 - 2023 ISSCC的Session 6: Advanced Wireline Links and Techniques
 - 2022 ISSCC的Session 6: Ultra-High-Speed Wireline 和 Session 17: Advanced Wireline Links and Techniques